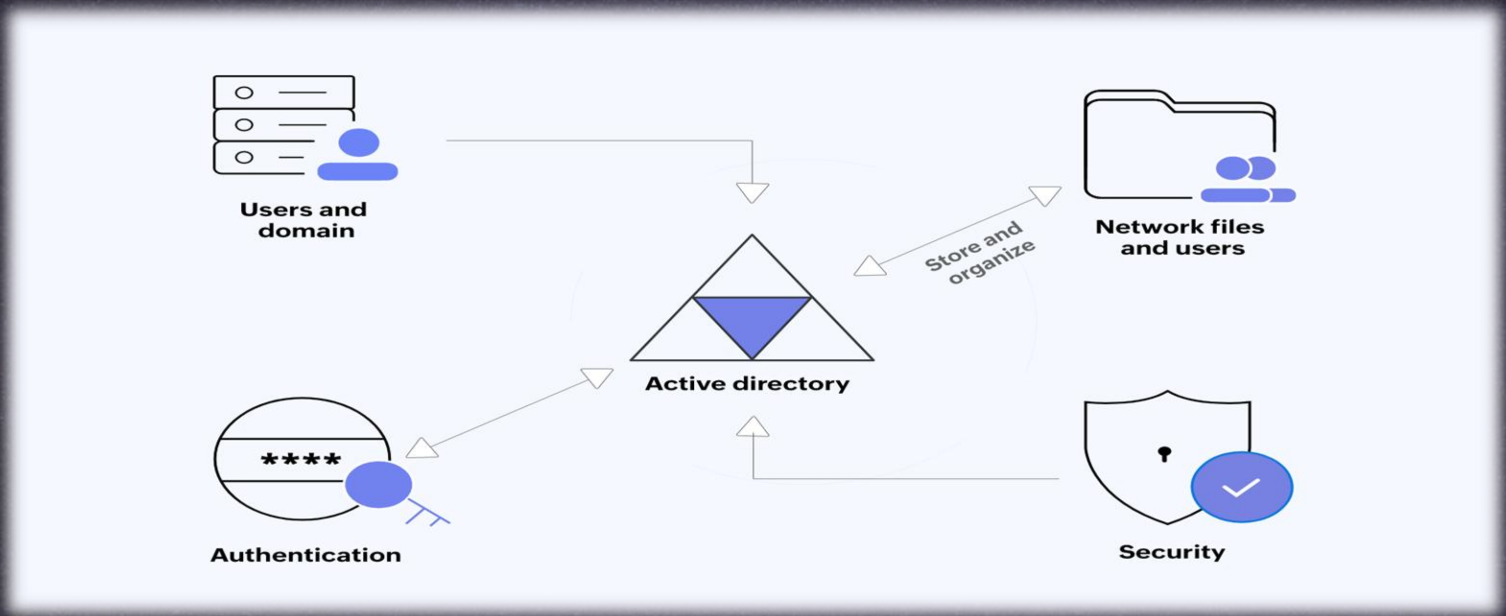
Module 02
Active Directory



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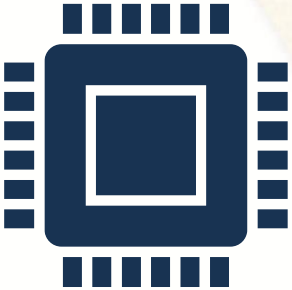
1. Introduction to Active Directory

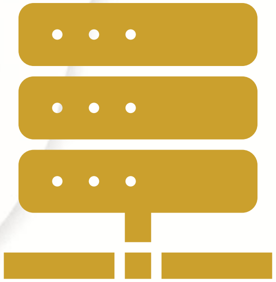
1.1. Domain Controller

A **Domain Controller**, commonly referred to as a **DC**, is simply a server that is hosting **Active Directory**. It is a central point of contact, a central "hub" so to speak, that is accessed prior to almost any communication that takes place between a client and server in your network.

Perhaps the easiest way to describe it is as a storage container for all identification that happens on the **network. Usernames, passwords, computer accounts, groups of computers, servers, groups and collections of servers, security policies, file replication services**, and many more things are stored within and managed by DCs.

1.1. Domain Controller (Cont.)





Domain Controller is a server that monitors and manages domains such as providing authentication and authorization. Authentication is the setting of permissions for clients to log on or not. Authorization is the setting of permissions for clients to log on and what they have the right to do.

A domain is a collection of devices such as Servers, Computers, Printers, Routers, Users, … to connect to a server-based network of an organization or a company. We can say that one network is equal to one domain and in each domain, there may be one or more Domain Controllers. Each domain must have a logical name, and the domain name should always be patterned after the organization name, for example: ***makara.com***

1.2. Active Directory

To become a server that can manage and provide services to clients, we need to make it a Domain Controller first. So, if we want to set up a **Domain Controller**, we need to set up Active Directory first, then it provides us with a Domain Controller as well.

Active Directory is a database used to centrally manage and store all information from a network or from an organization's domain. Alternatively, it is a database that stores information from network objects, such as computers, printers. , Users, Applications, Security Policy, database, ...

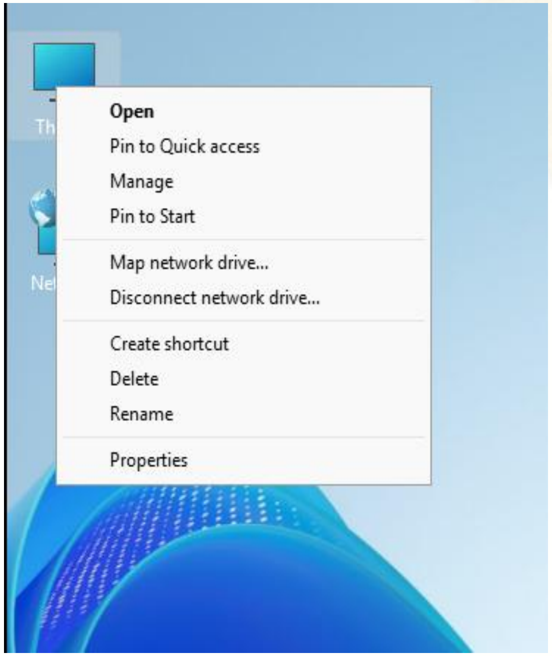
2. Prepare before Installing the AD DS role

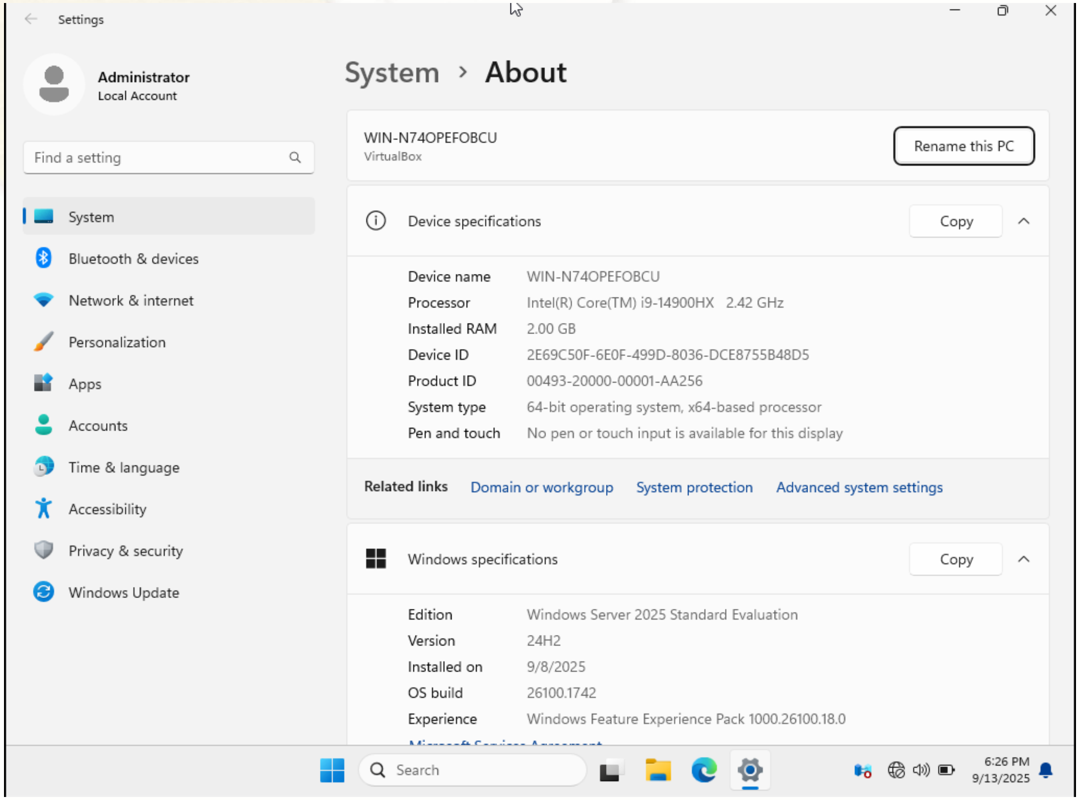
2.1. Server Requirements

Set a static IP address	192.168.1.2/24
Set a good hostname	WIN-SERVER
Set a domain name	makara.com
Set a DNS server address	192.168.1.2

2.2. Change Hostname

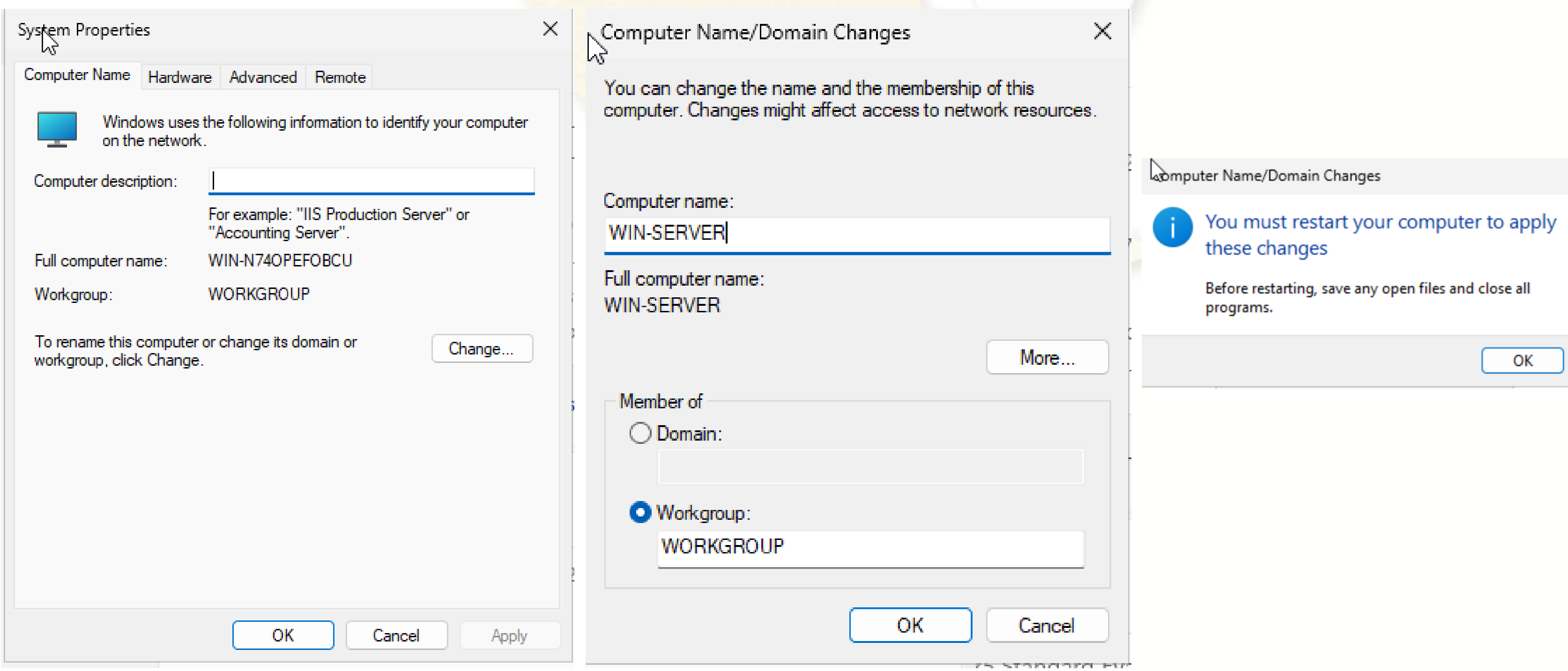
- Step 1: Right click on **This PC** > Choose **Properties** > Click **Domain or workgroup** link





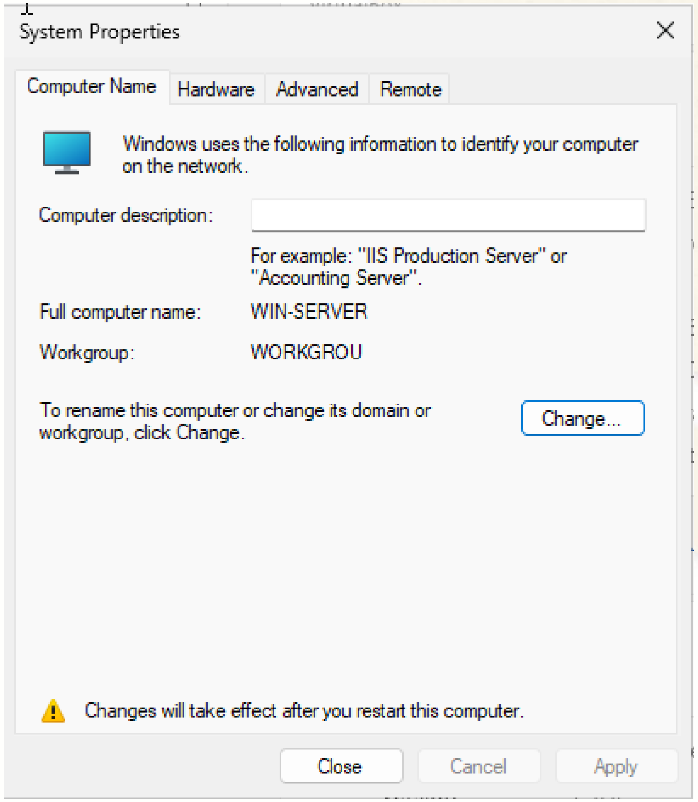
2.2. Change Hostname (Cont.)

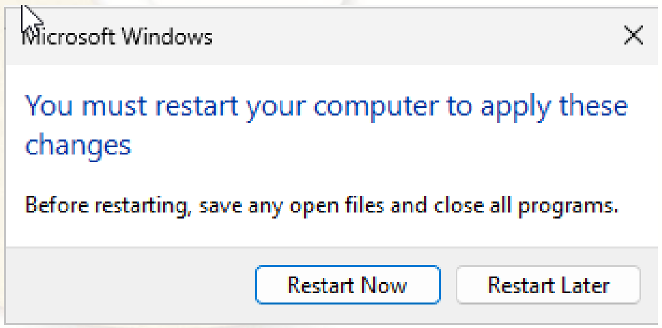
- Step 2: Click **Change** button > In **Computer name:** Enter new hostname **(WIN-SERVER)** > Click **OK** > Click **OK**



2.2 Change Hostname (Cont.)

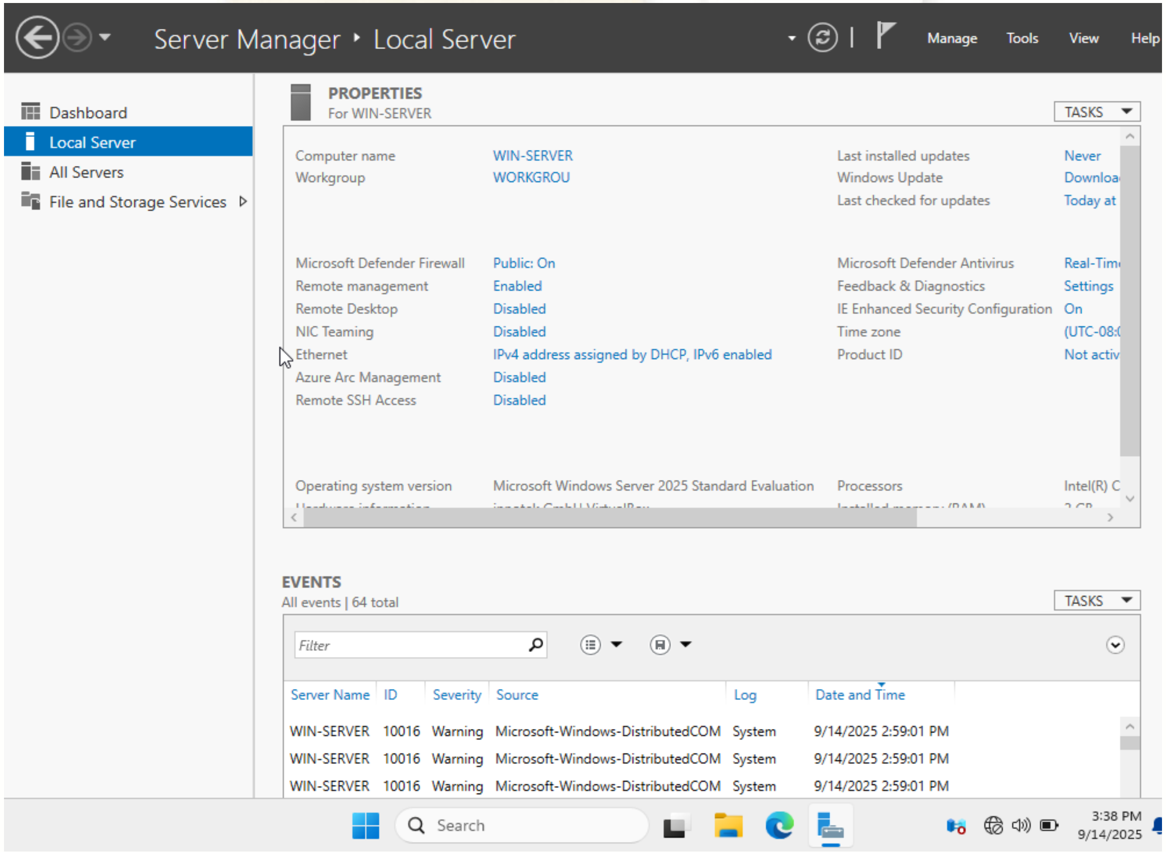
- Step 3: Click **Close** > Click **Restart Now** button





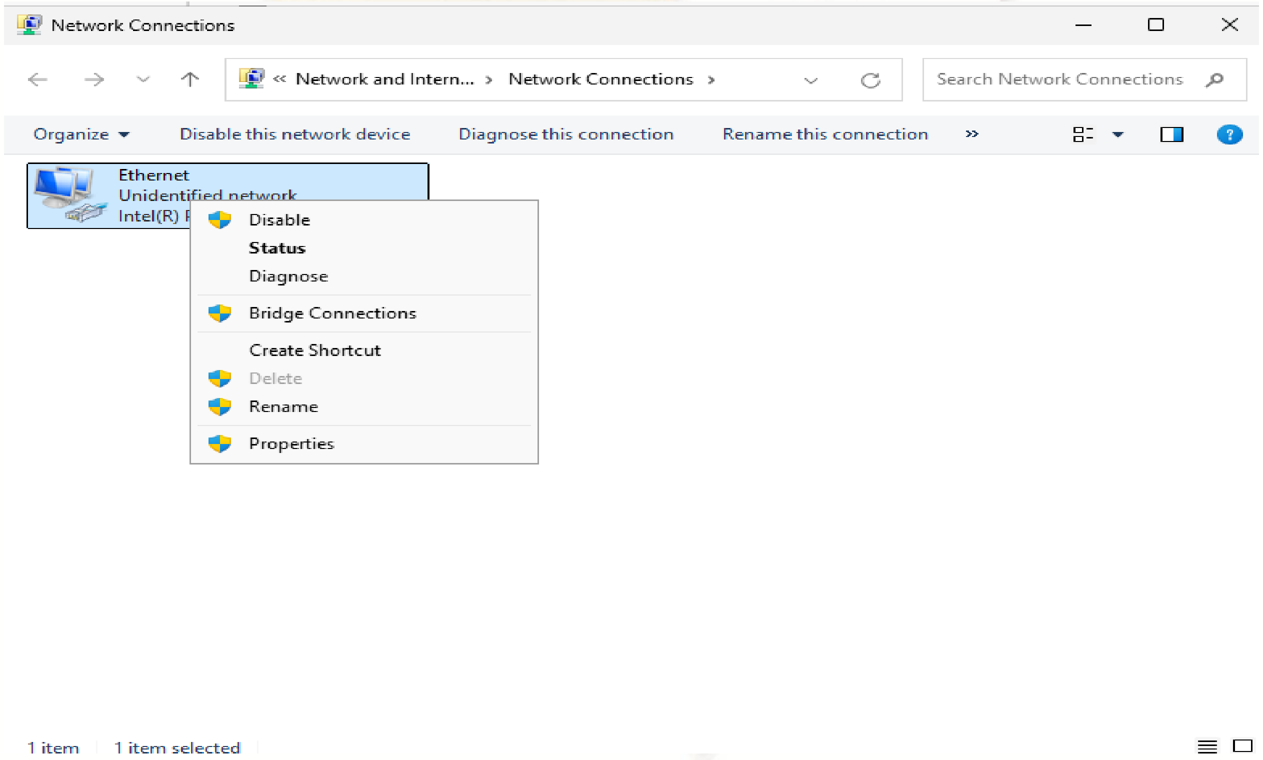
2.3. Set IP Address

- Step 1: Click on **<u>Local Server</u>** in **Server Manager** icon > Choose on Ethernet **<u>IPv4 address assigned by DHCP, IPv6 enabled</u>**



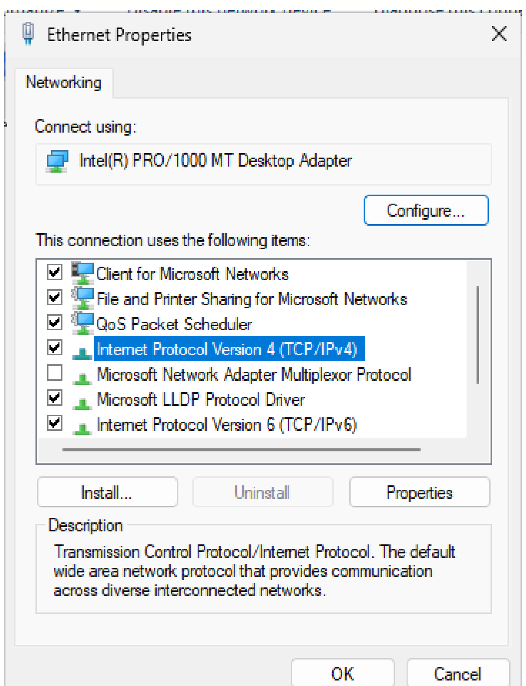
2.3. Set IP Address (Cont.)

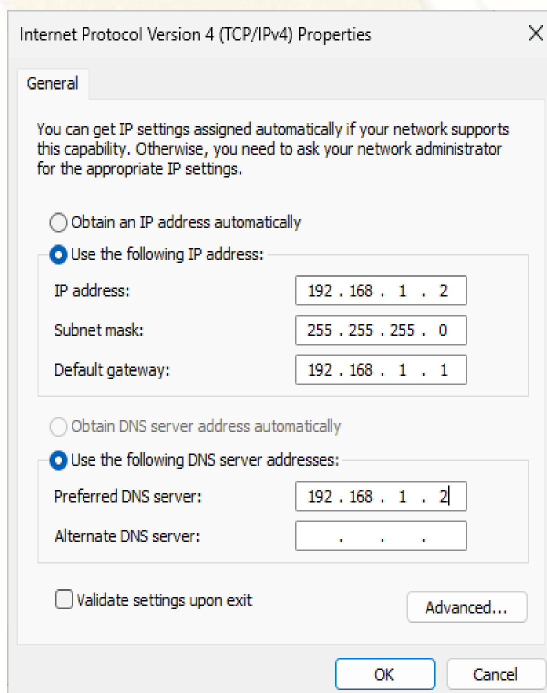
- Step 2: Right click on **Network Adapter** (Ethernet) > Choose **Properties**

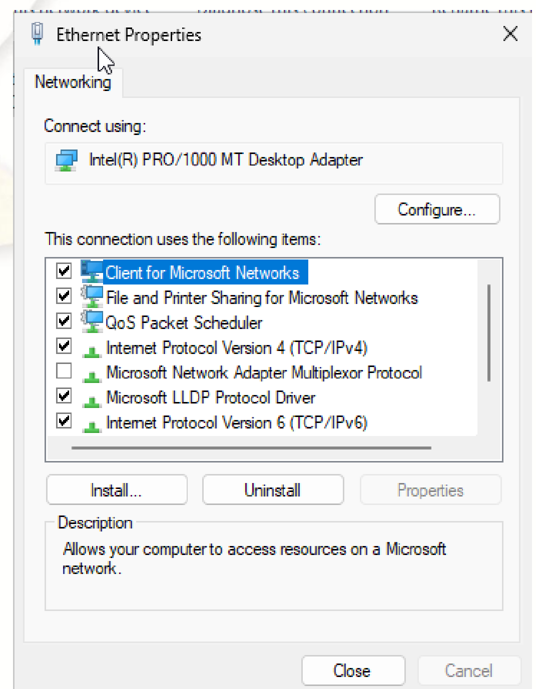


2.3. Set IP Address (Cont.)

- Step 3: Select **Internet Protocol Version 4 (TCP/IPv4)** > Click **Properties** button > Enter **IP address, Subnet mask, Default gateway** and **Preferred DNS server** > Click **OK** > Click **Close**



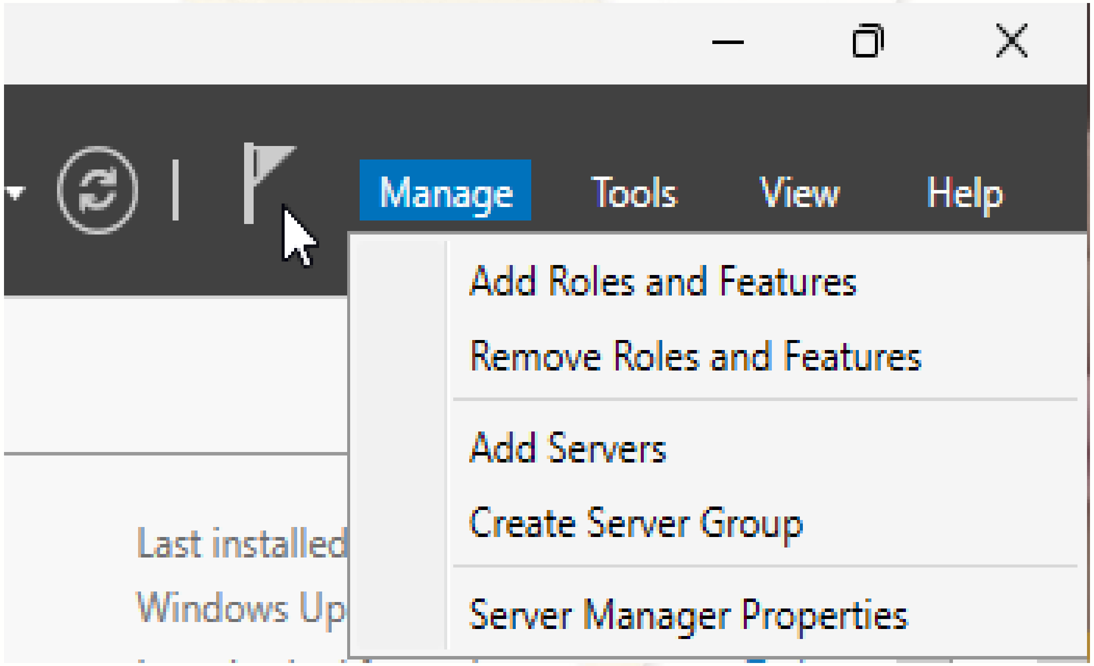




3. Install the AD DS role

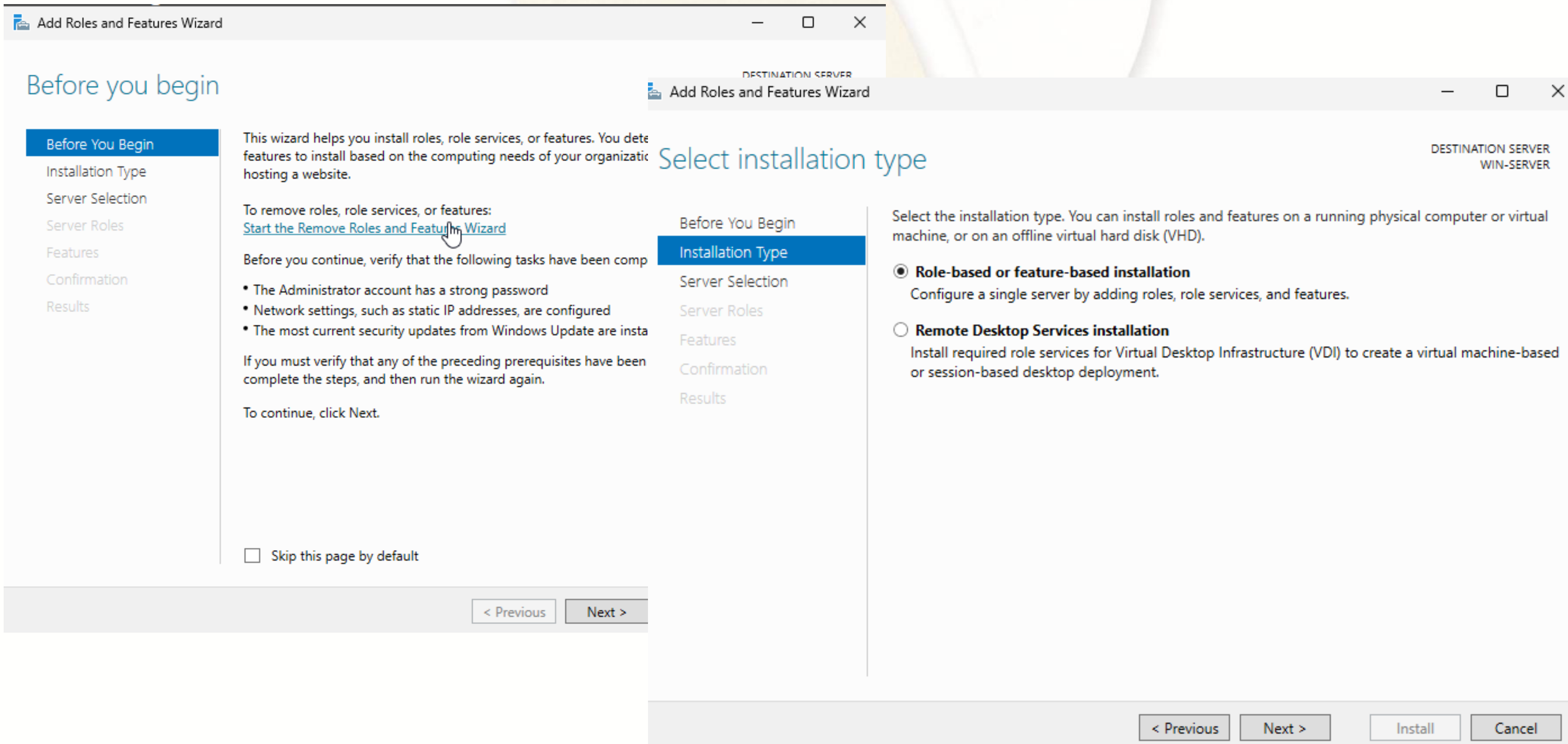
3.1. Installation

- Step 1: Click **Manage** menu > Click **Add Roles and Features**



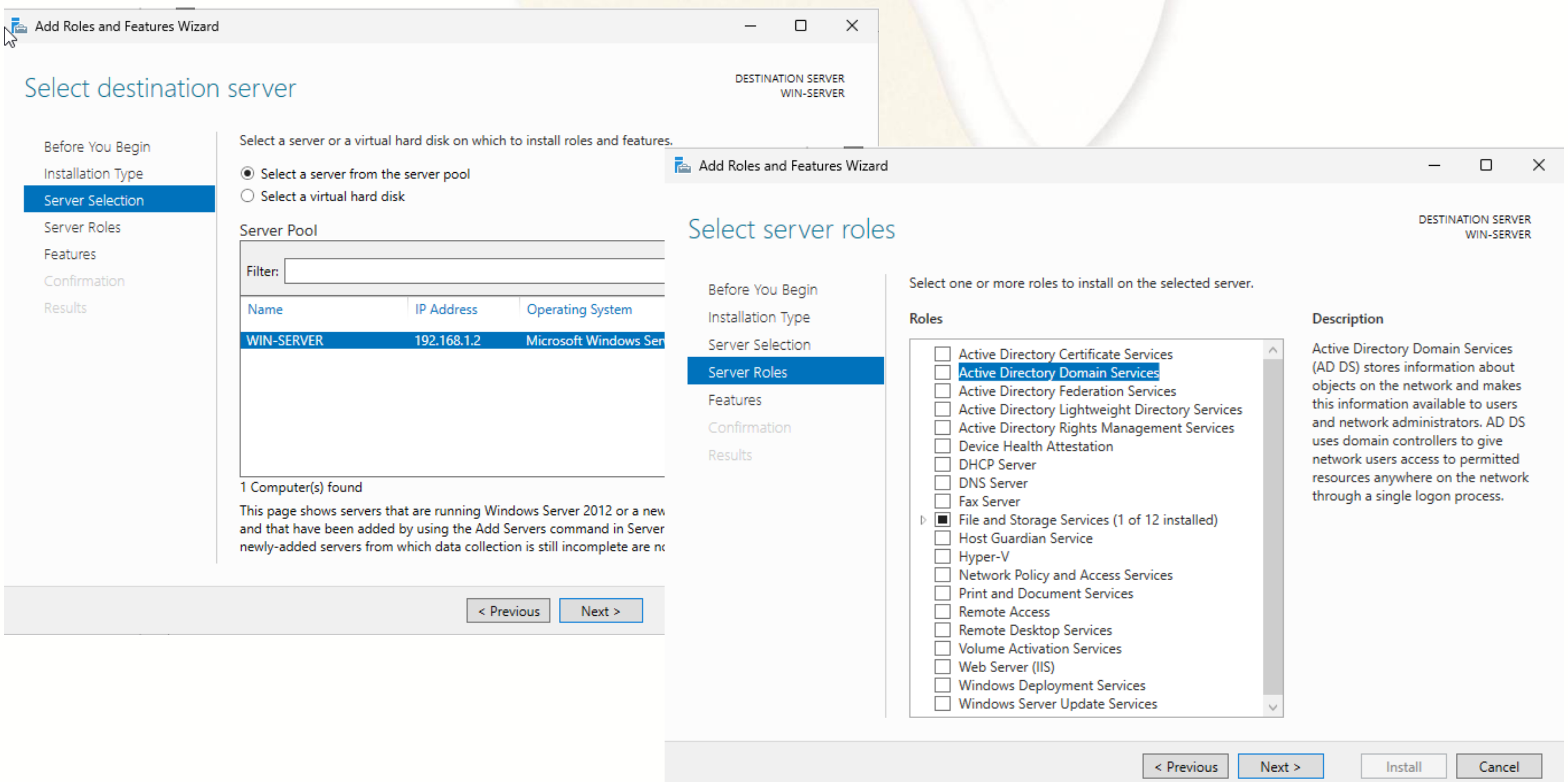
3.1. Installation (Cont.)

- Step 2: On **Before you begin** window, click **Next** button > On **Select installation type** windows, click **Next** button >



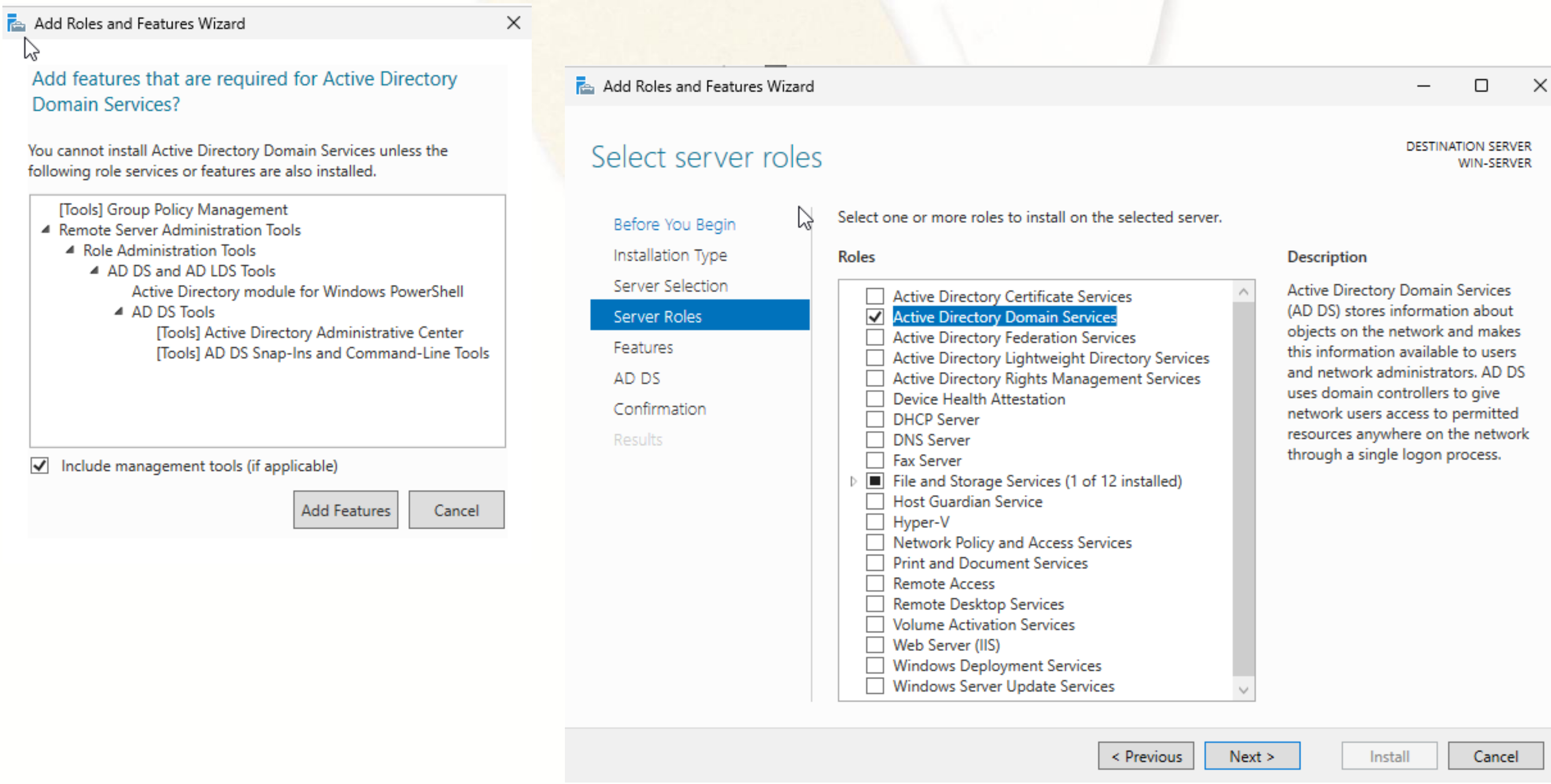
3.1. Installation (Cont.)

- Step 3: On **Select destination server**, click **Next** button >On **Select server roles**, check **Active Directory Domain Services**



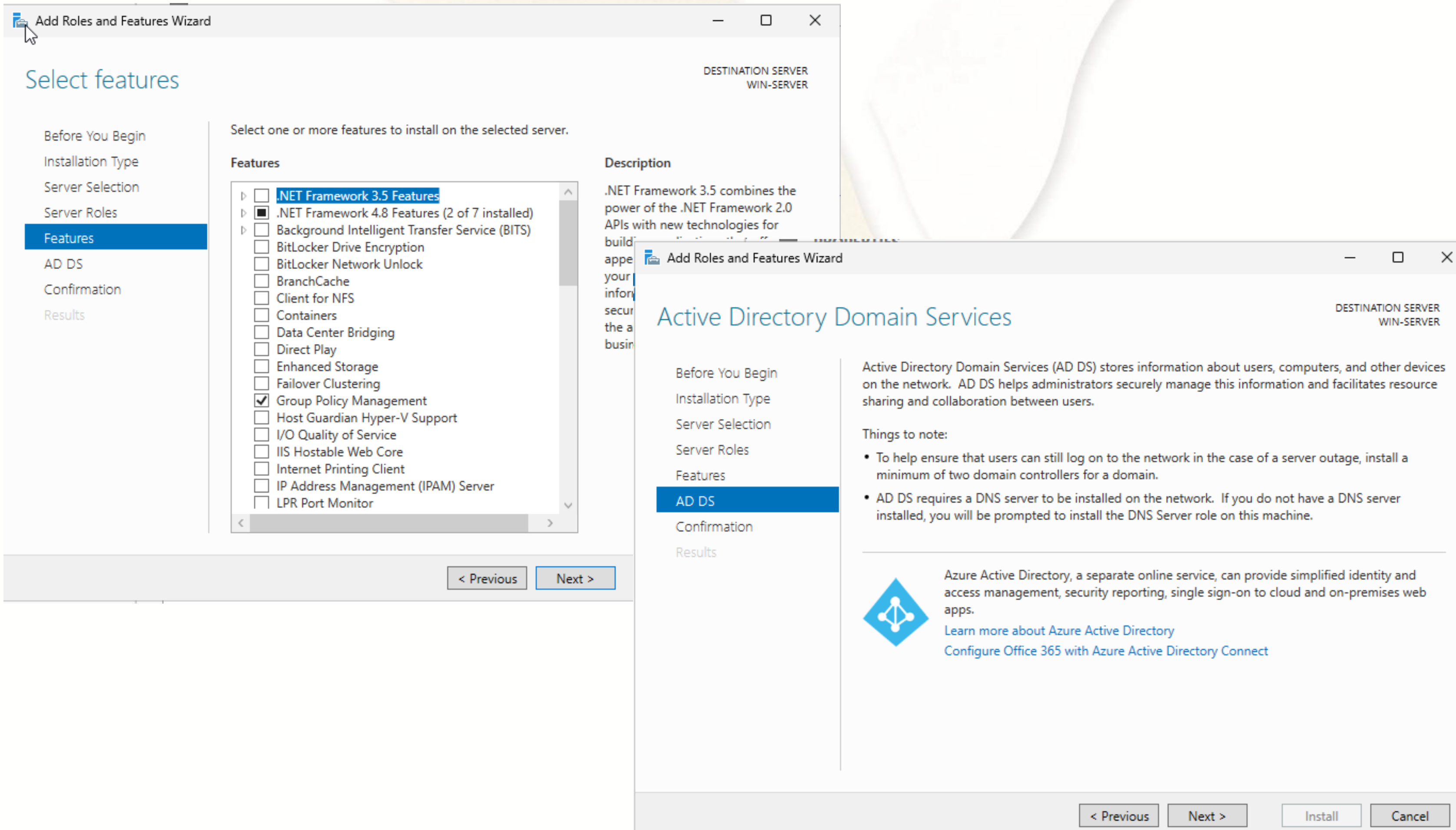
3.1. Installation (Cont.)

- Step 4: Click **Add Features** button > Click **Next**



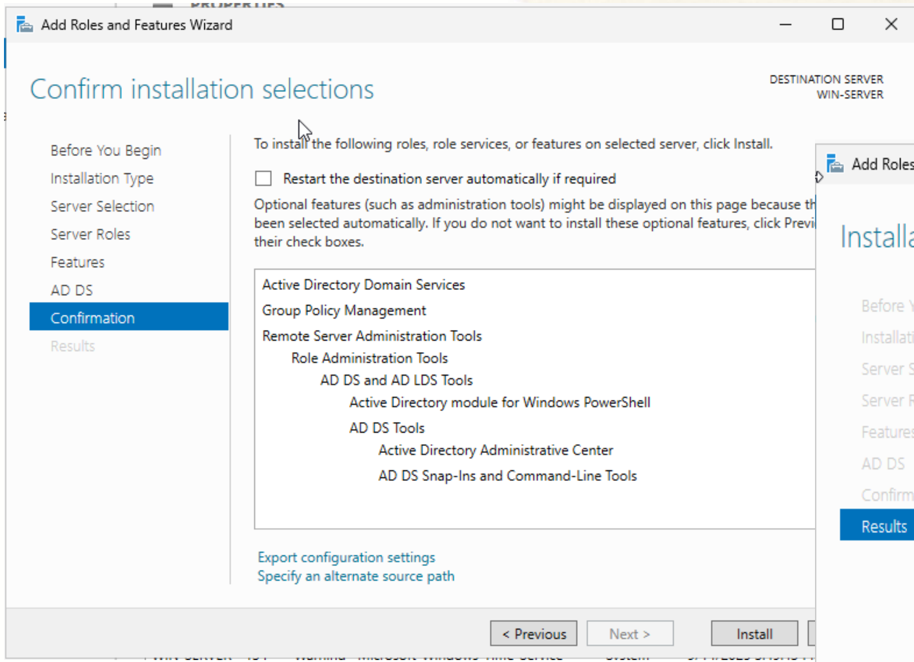
3.1. Installation (Cont.)

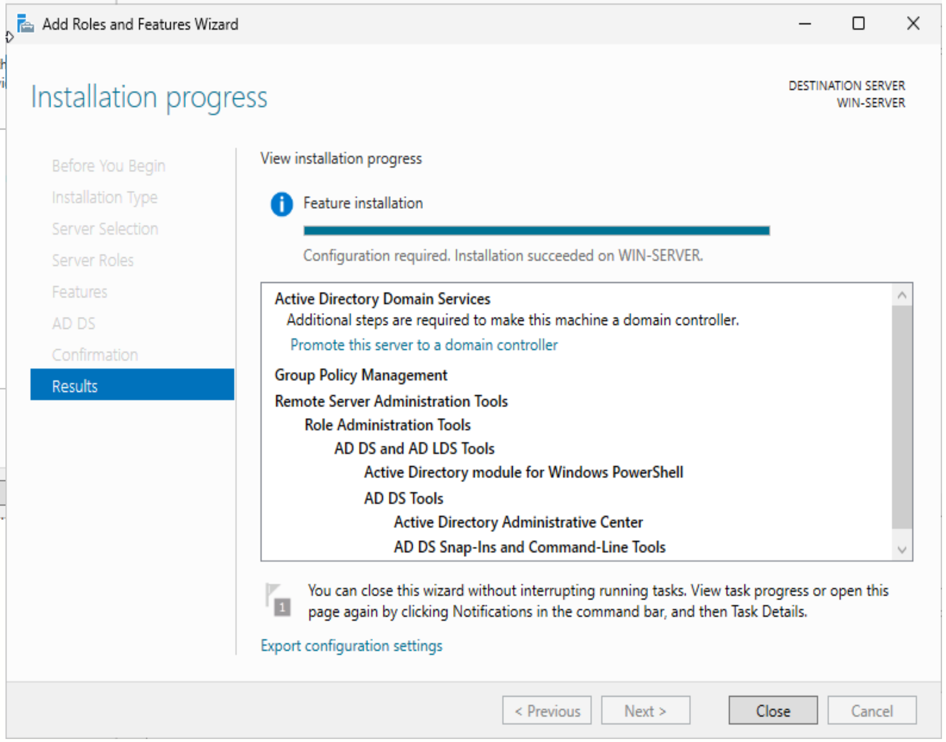
- Step 5: Click **Next** > Click **Next**



Installation (Cont.)

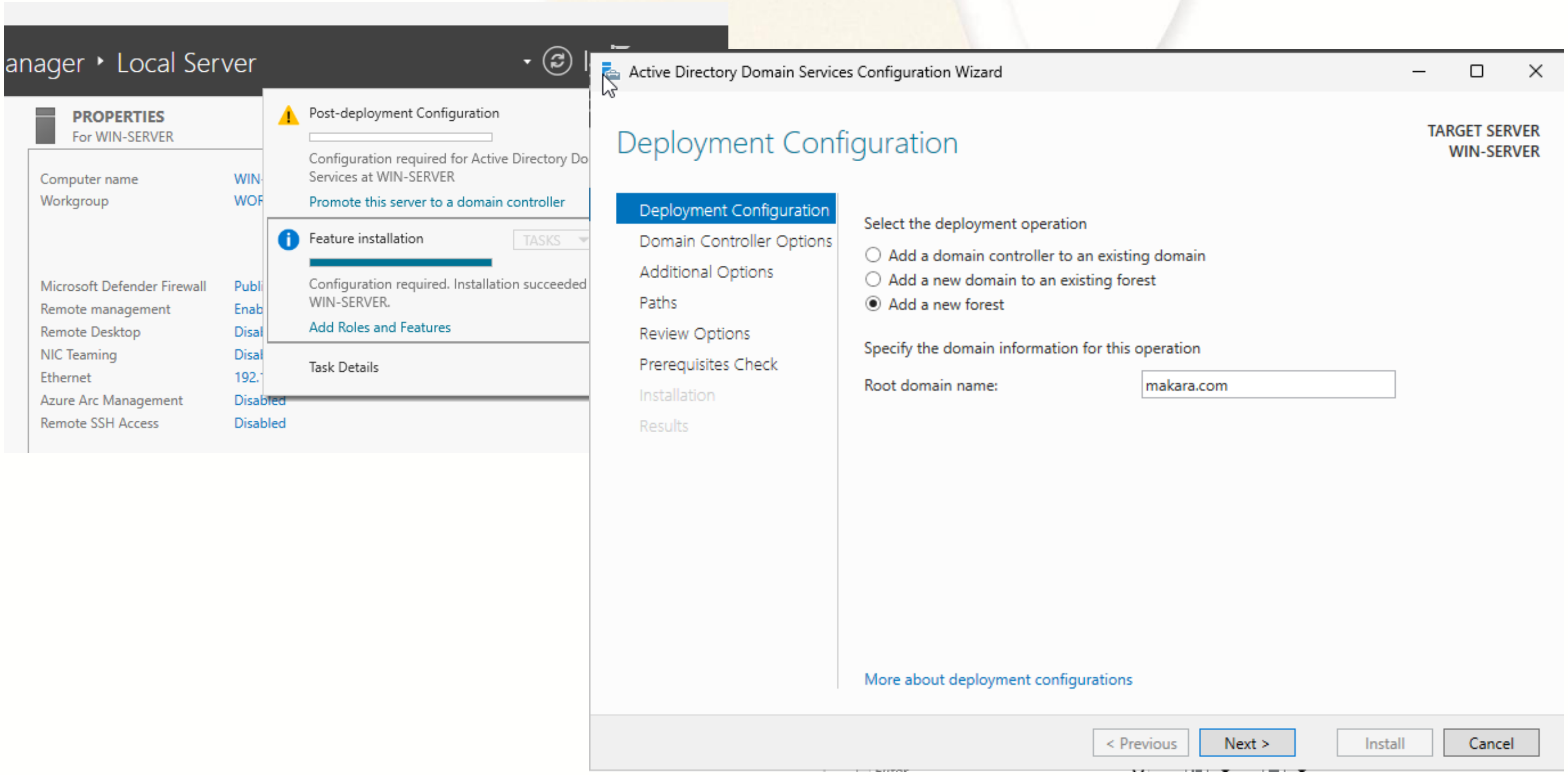
- Step 6: Click **Install** > *Wait until installation succeeded*, Click **Close**





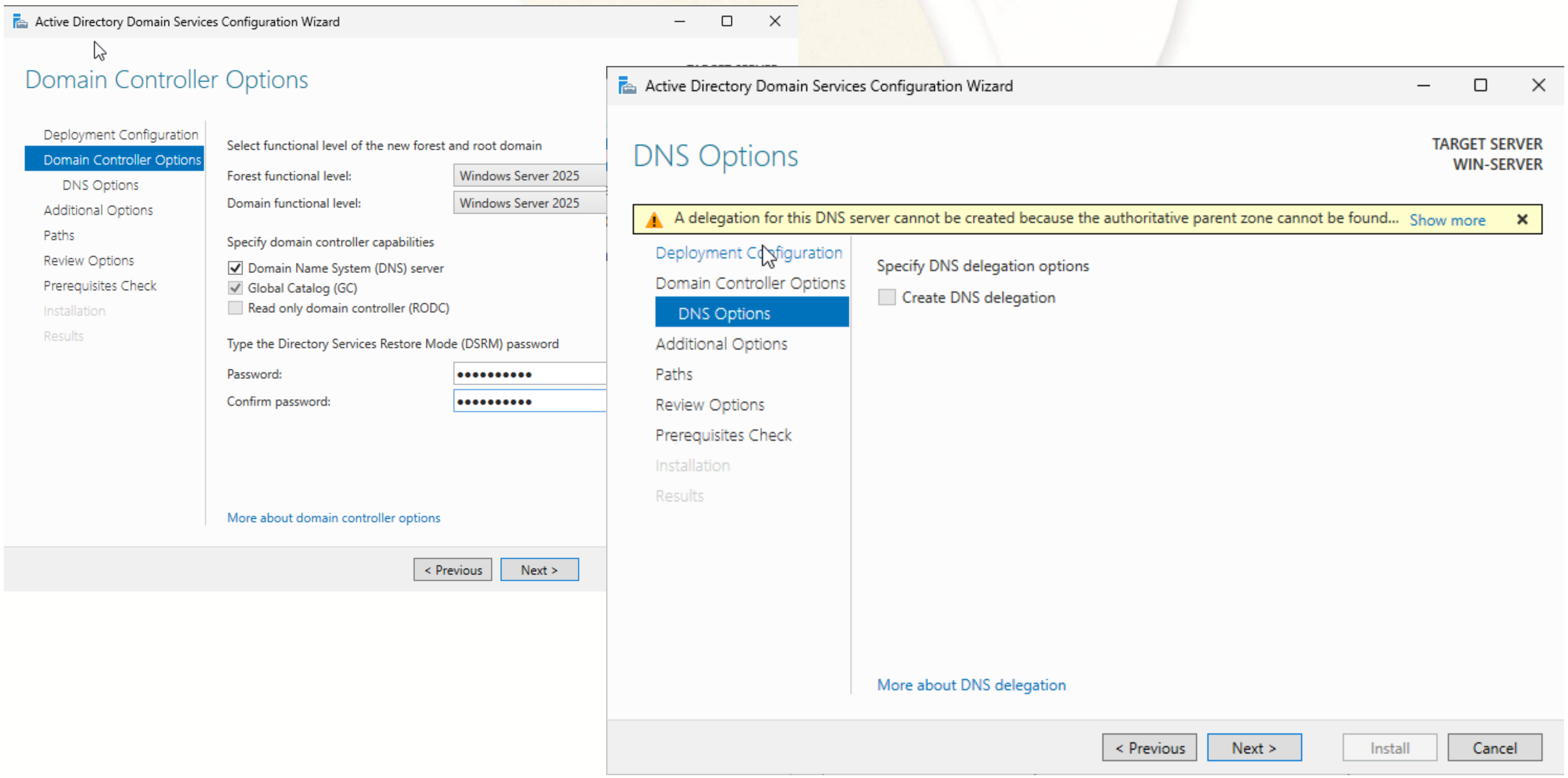
3.1. Installation (Cont.)

- Step 7: Click **Notification menu** > Click **Promote this server to a domain controller** link > Select **Add a new forest** > Enter Root **domain name: (makara.com)** > Click **Next**



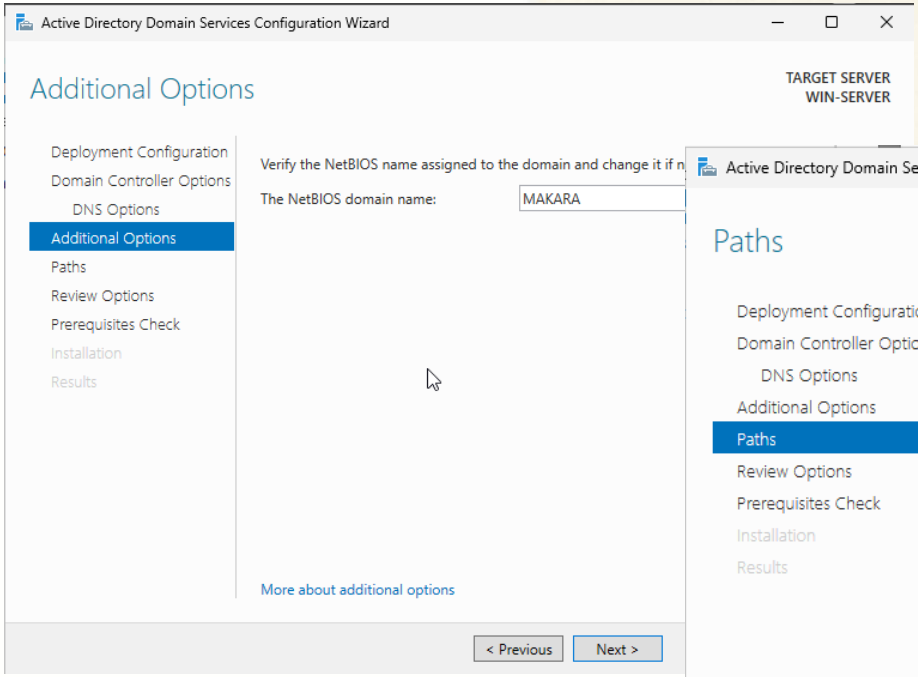
3.1. Installation (Cont.)

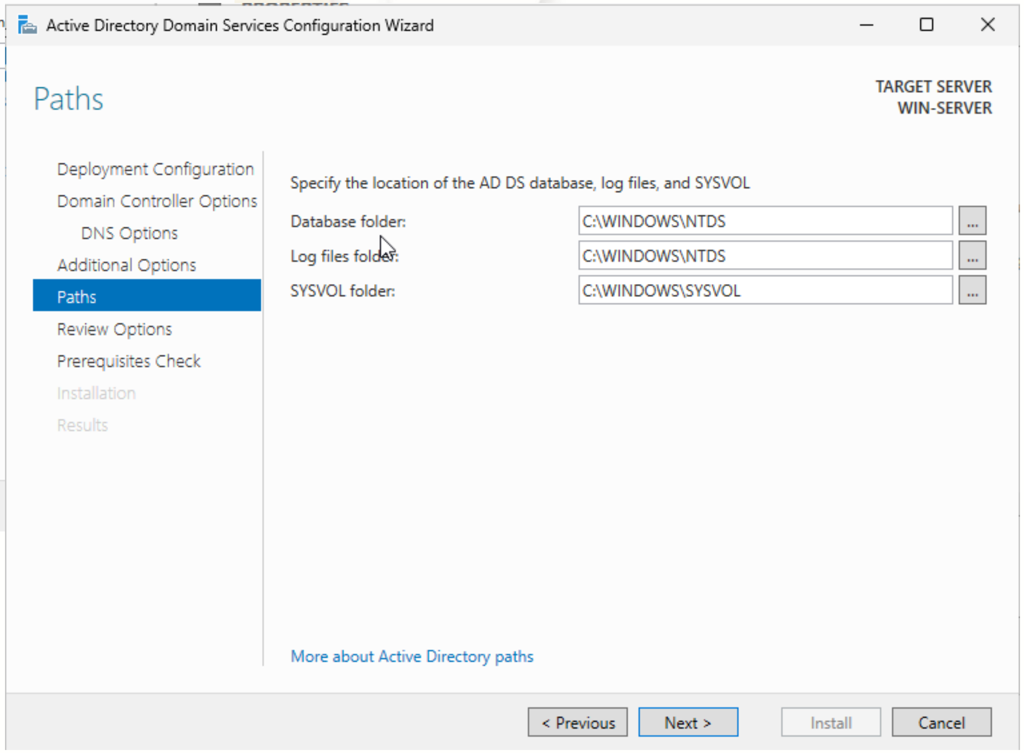
- Step 8: Enter password twice in **Password** and **Confirm password** text fields **(for recovery process)** > Click **Next** > Click **Next**



3.1. Installation (Cont.)

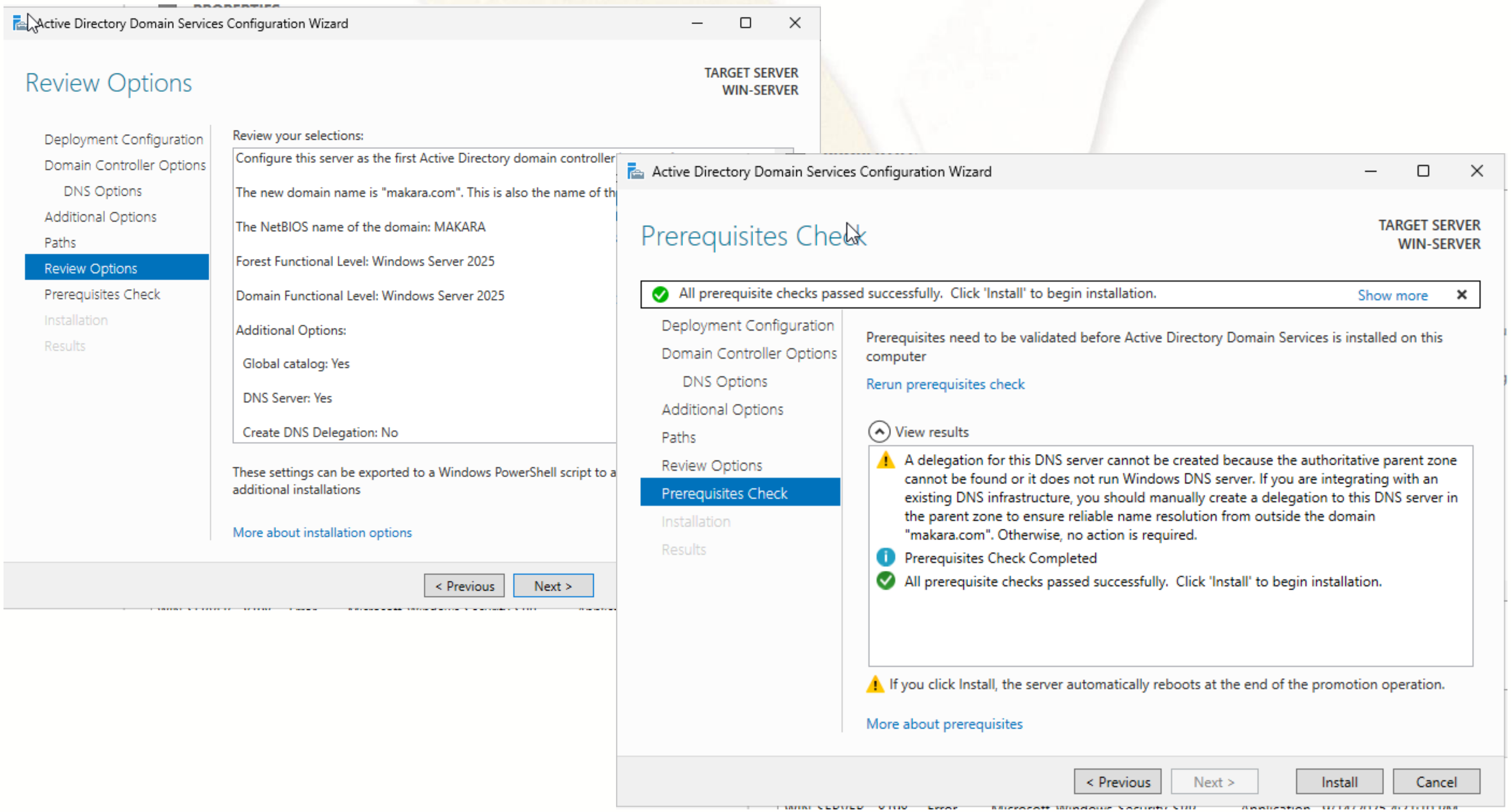
- Step 9: The **NetBIOS domain name** will be generated by itself, just click **Next** button > Click **Next**





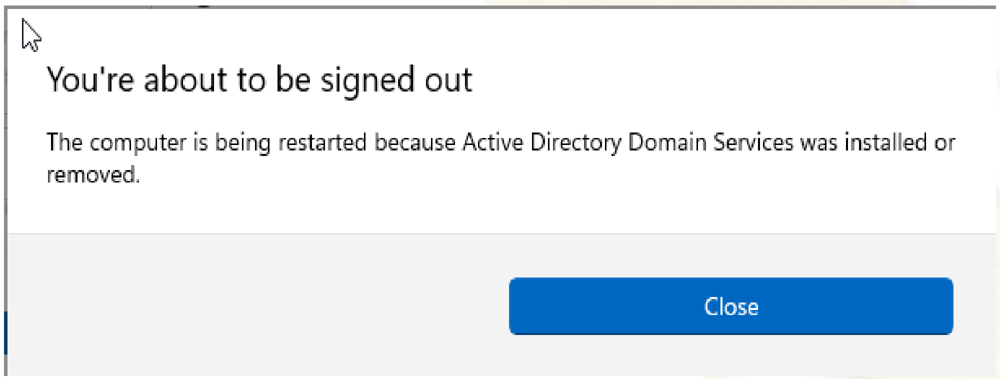
3.1. Installation (Cont.)

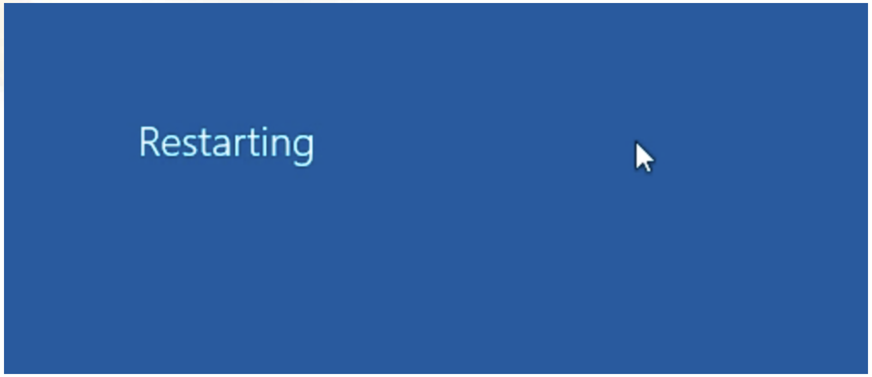
- Step 10: Click **Next** > Click **Install**



3.1. Installation (Cont.)

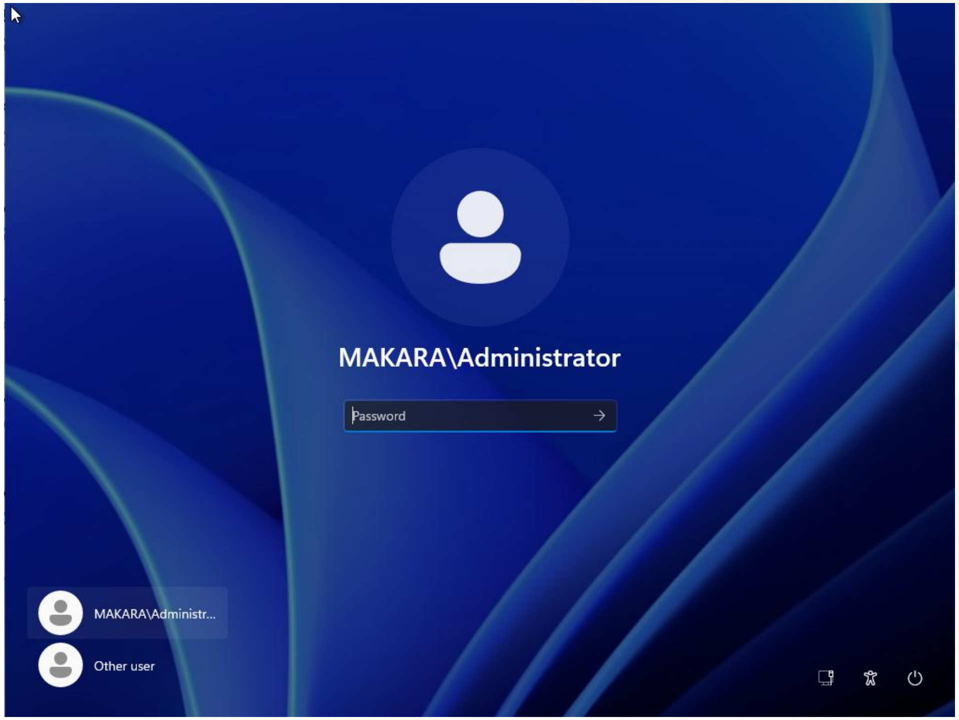
- Step 11: Server will **restart by itself**, just wait until it **starts up**





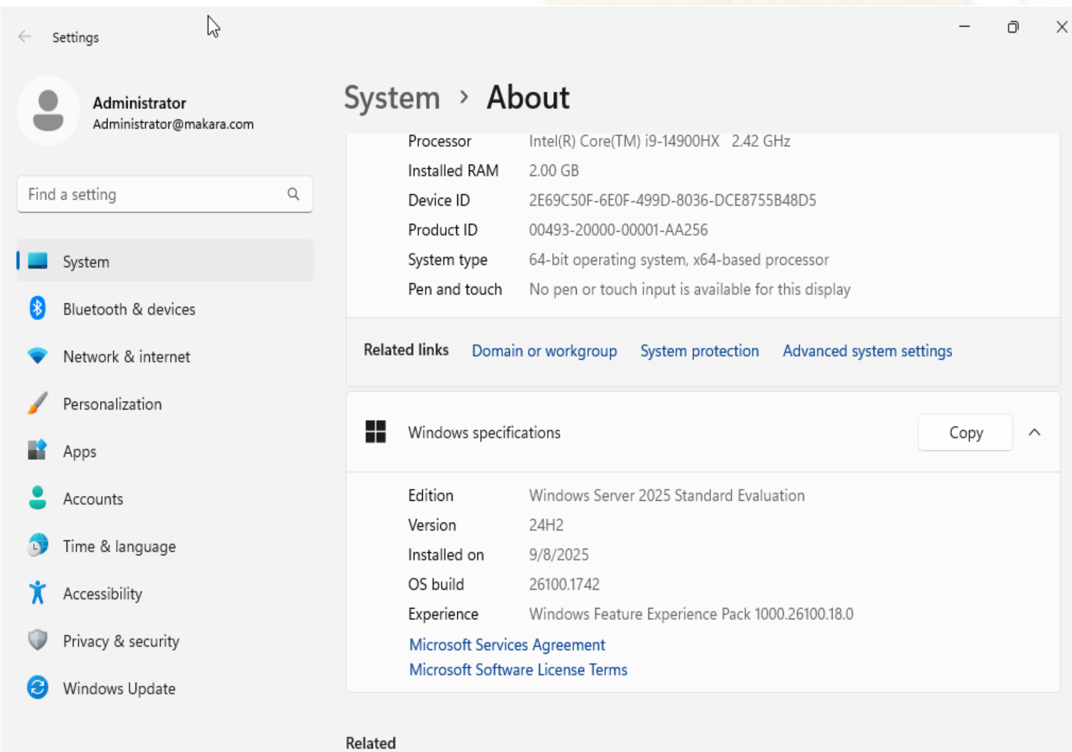
3.1. Installation (Cont.)

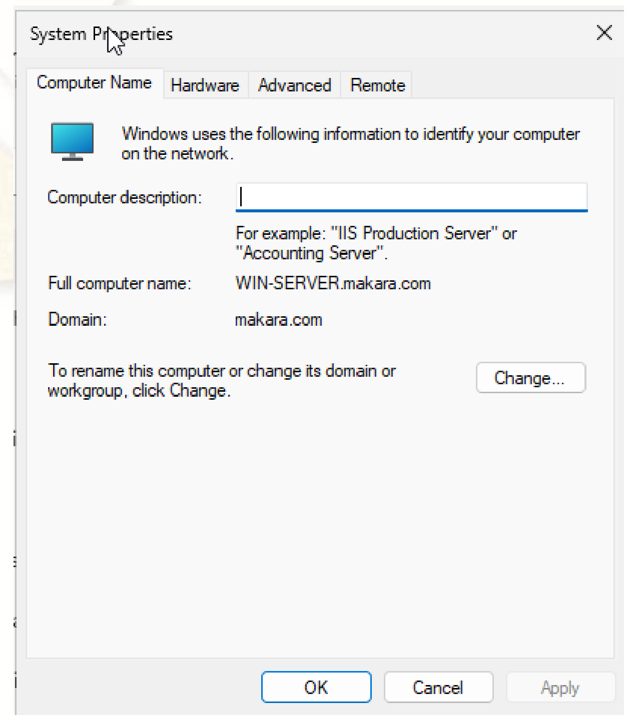
- Step 12: Enter **username** (MAKARA\Administrator) and **password** to login > Right click on **This PC** > Select **Properties**



3.1. Installation (Cont.)

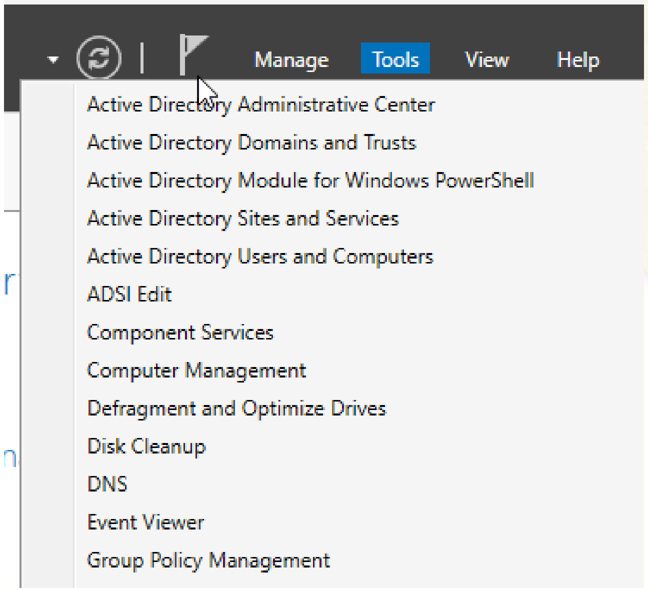
- Step 13: Click on <u>**Domain or workgroup**</u> you will see Full computer name: **WIN-SERVER.makara.com** and on **Domain: makara.com**

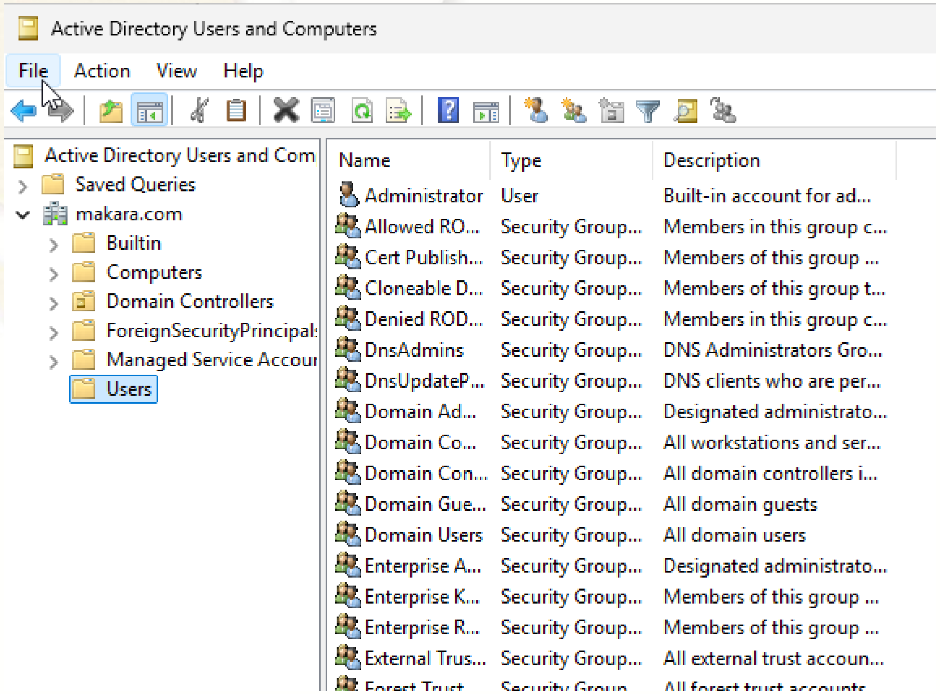




3.2. Create a User

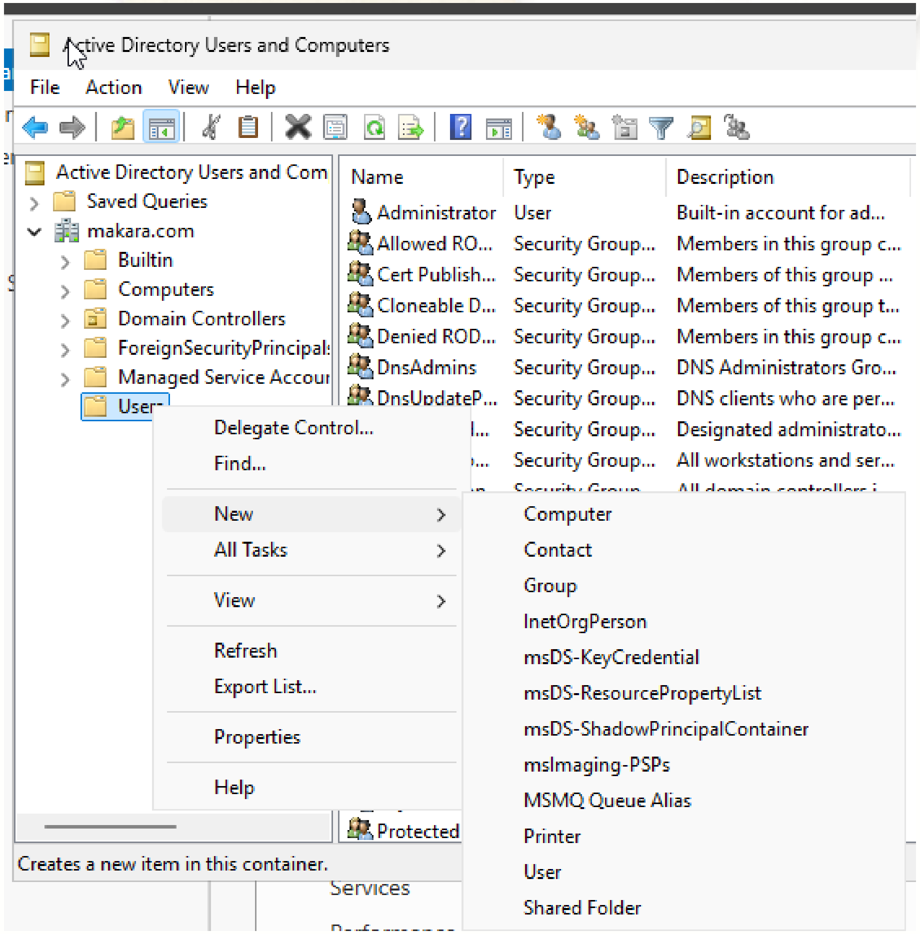
- Step 1: Click **Tools** menu > Click **Active Directory Users and Computers** > Expand **domain name (makara.com)** > Select **Users** Folder





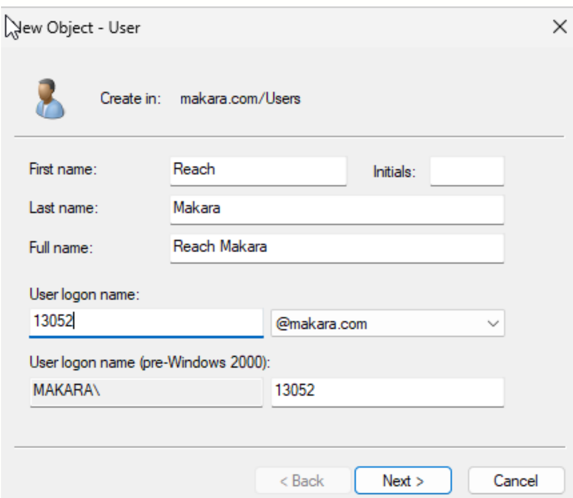
3.2. Create a User (Cont.)

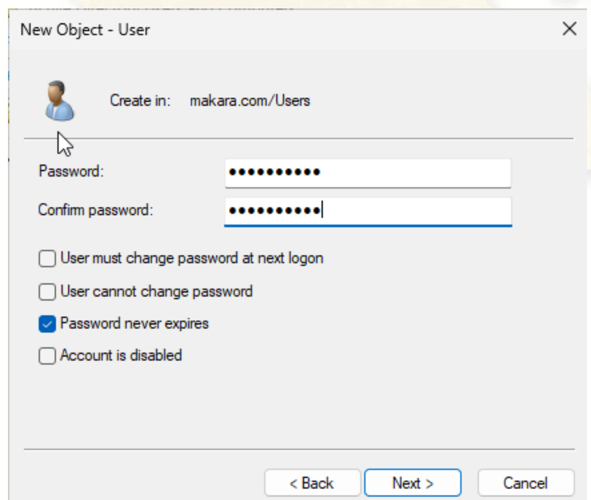
- Step 2: Right click on **user list** > **New** > **User**

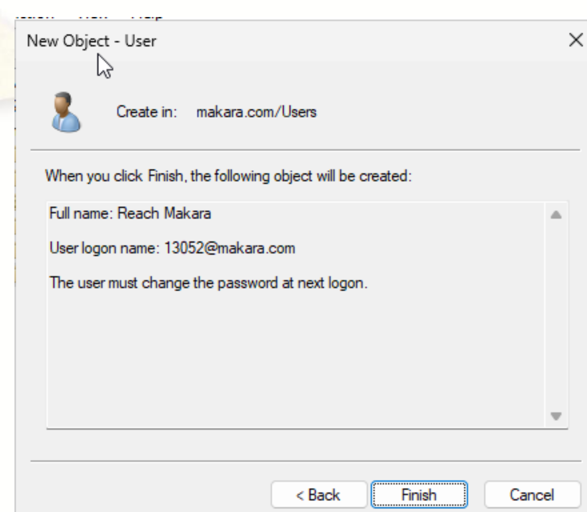


3.2. Create a User (Cont.)

- Step 3: Enter user information in **First name**, **Full name**, **User logon name** > Click **Next** > Enter password twice in **Password** and **Confirm password** > Click **Next** > Click **Finish**

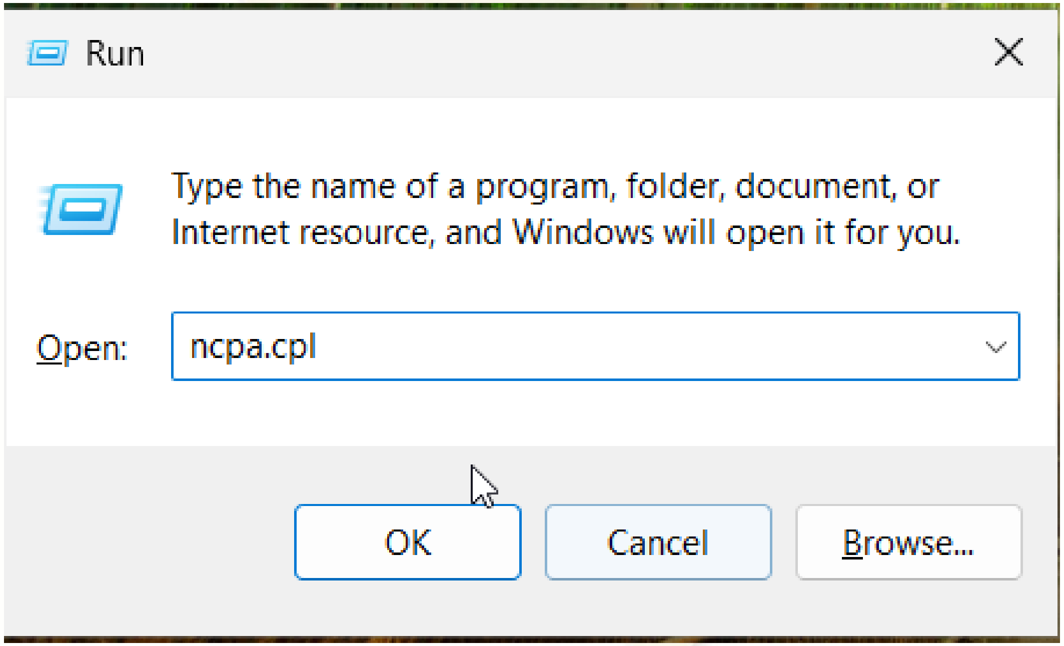






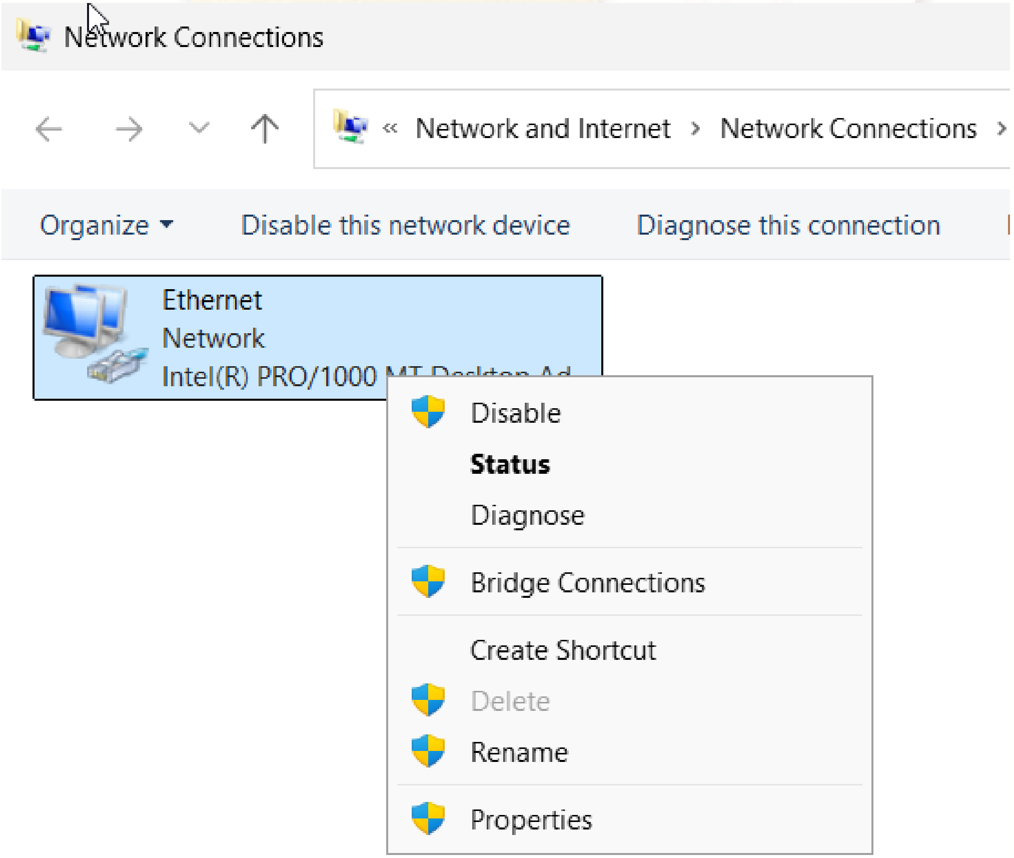
3.3. Join a Computer to a Domain

- Step 1: On **client computer**, go to **Run (Key Windows + R)** type **ncpa.cpl** > Click **OK or hit Enter**



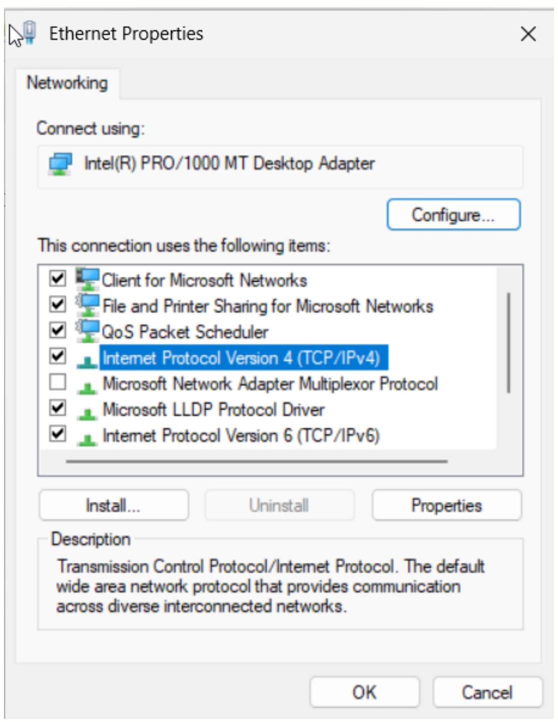
3.3. Join a Computer to a Domain (Cont.)

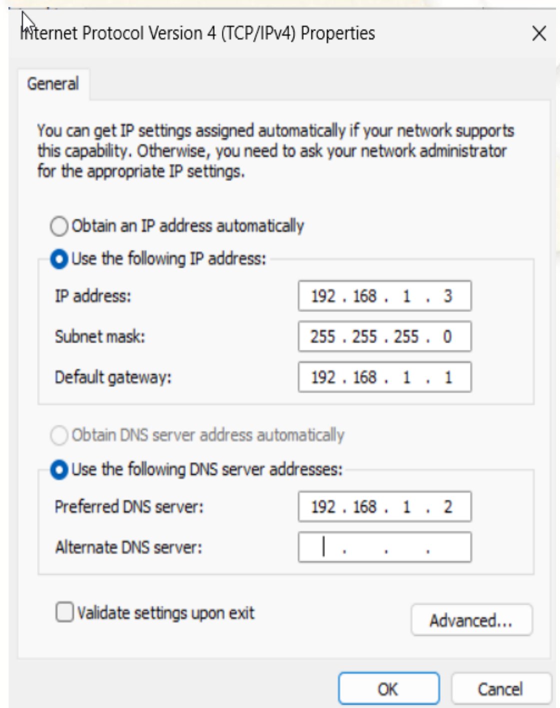
- Step 2: Right click on **network adapter** (Ethernet) > Select **Properties**

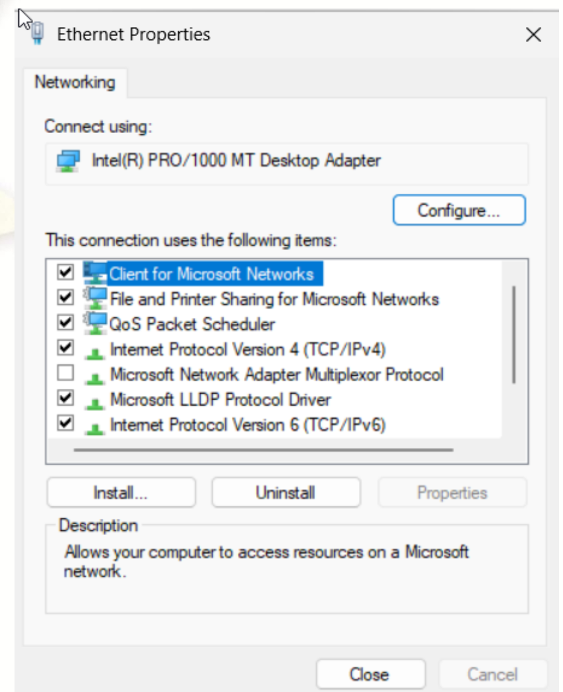


3.3. Join a Computer to a Domain (Cont.)

- Step 3: Select **Internet Protocol Version 4 (TCP/IPv4)** > Click **Properties** button > Enter **IP address**, **Subnet mask**, **Default gateway** and **Preferred DNS server** > Click **OK** > Click **Close**

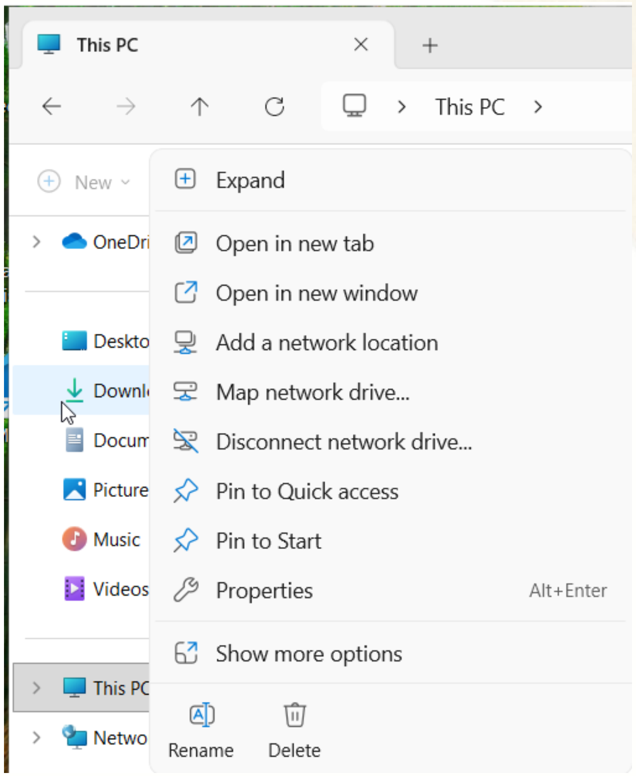


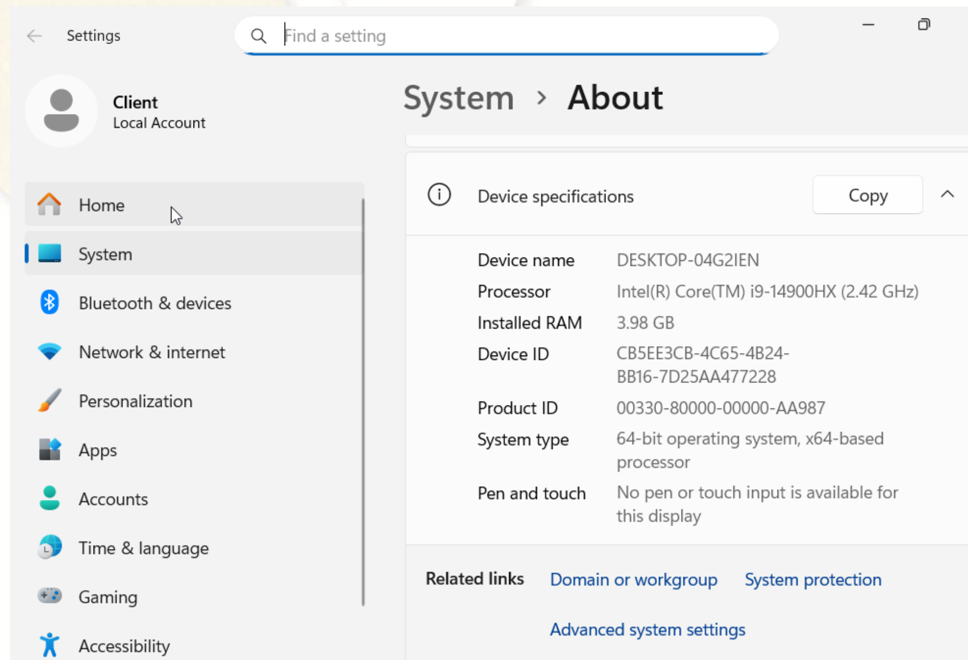




3.3. Join a Computer to a Domain (Cont.)

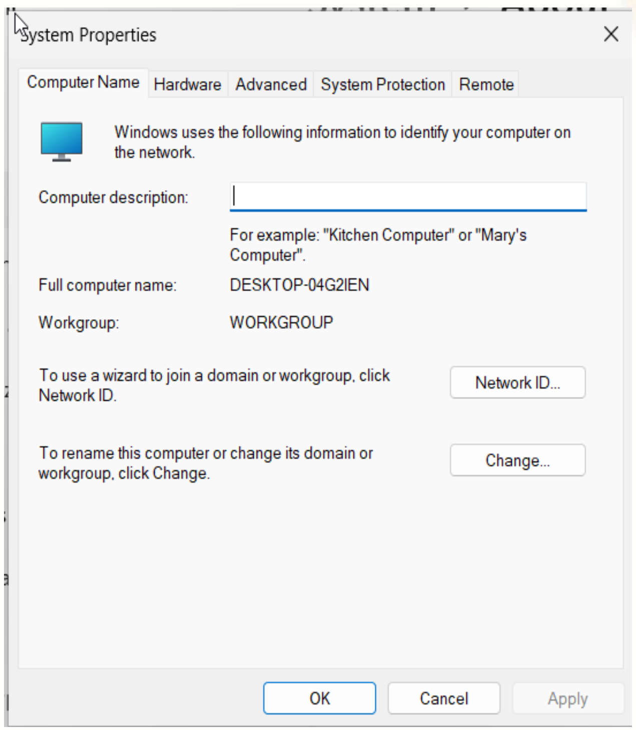
- Step 4: Right click on **This PC** > Select **Properties** > Click on **Domain or workgroup** link

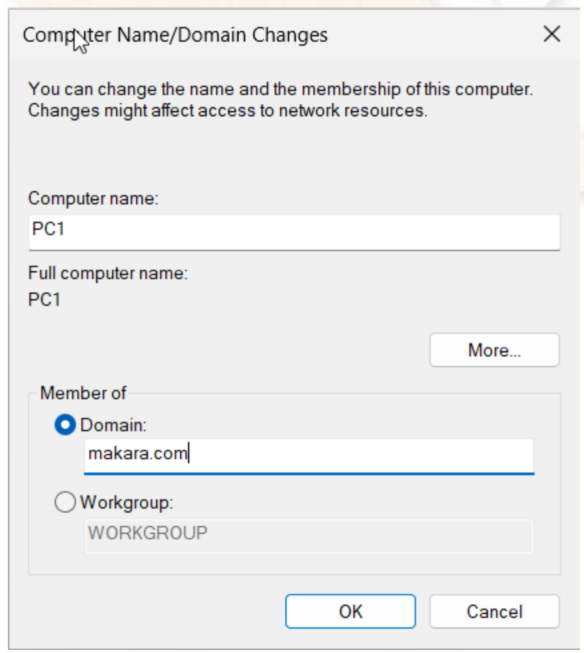


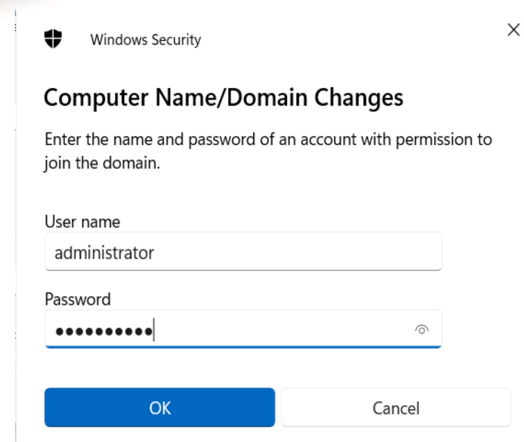


3.3. Join a Computer to a Domain (Cont.)

- Step 5: Click **Change** button > Select **Domain:** > Enter **domain name (makara.com)** > Click **OK** > Enter **Domain Administrator username and password** > Click **OK**

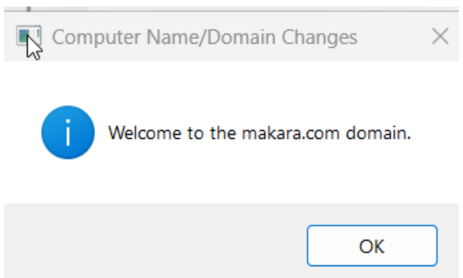


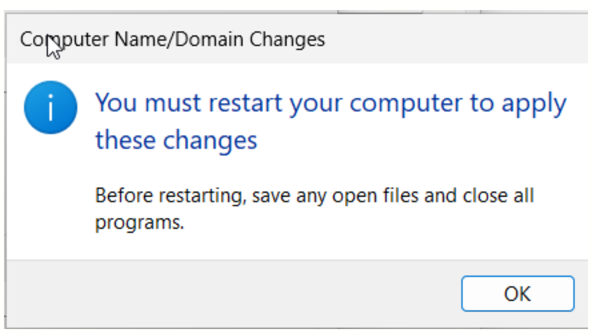


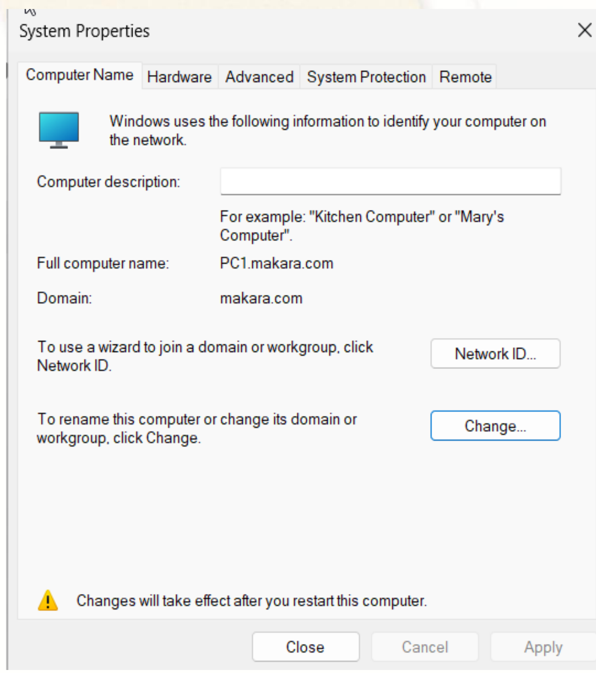


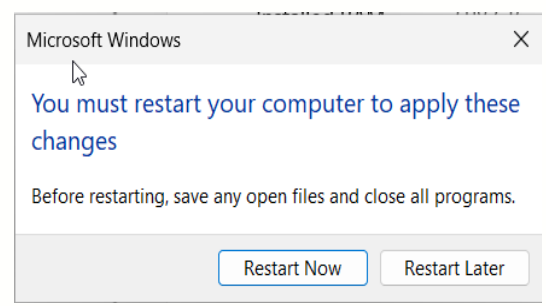
3.3. Join a Computer to a Domain (Cont.)

- Step 6: On **Computer Name/Domain Changes**, Click **OK** > Click **OK** > Click **Close** > Click **Restart Now** button



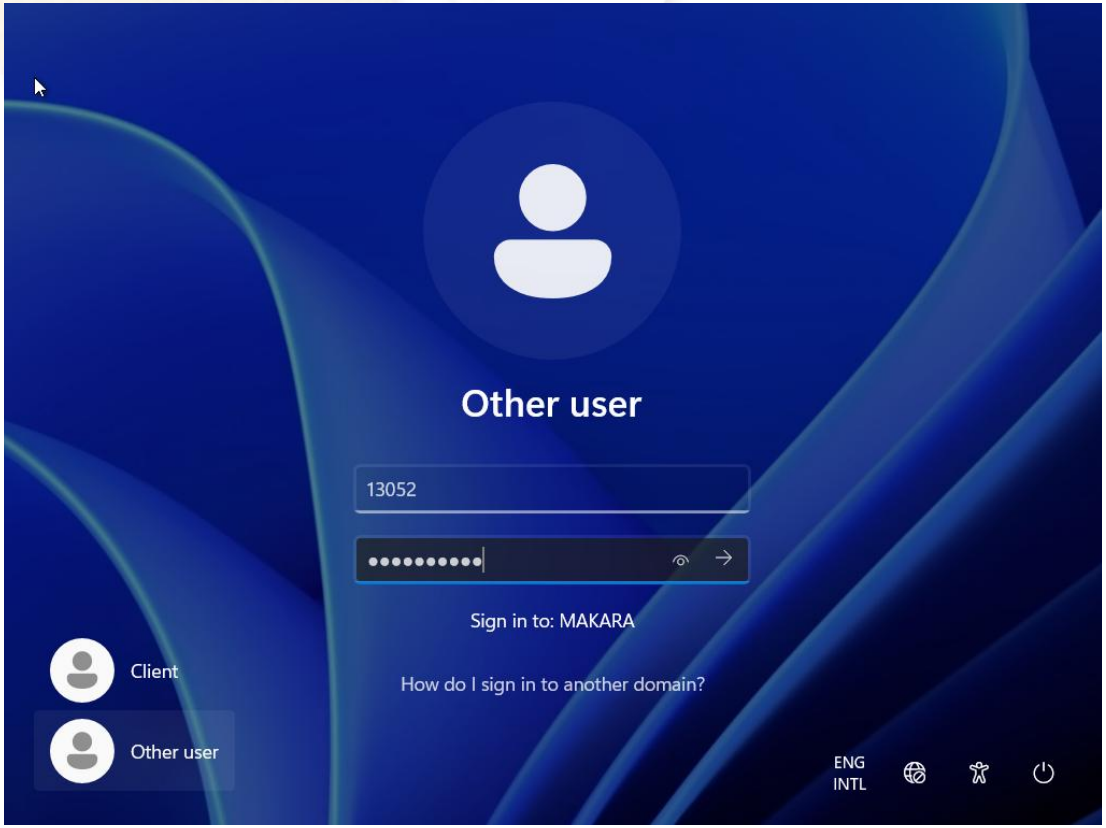






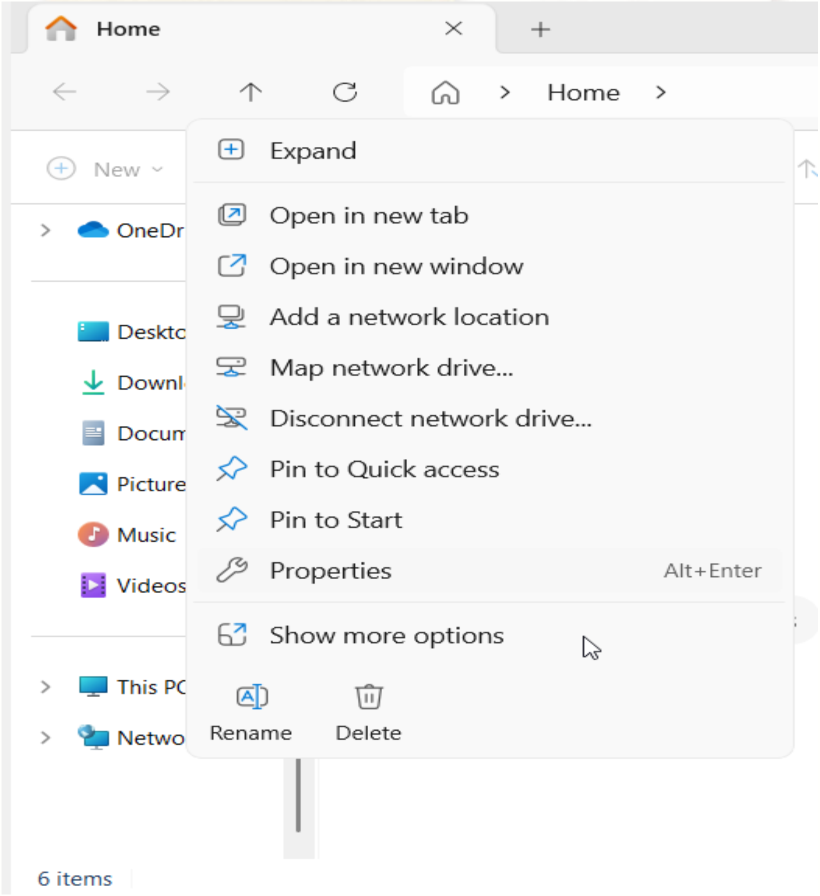
3.3. Join a Computer to a Domain (Cont.)

- Step 7: After **restart**, Click **Other user** > On **Username** please input **User** that create on Server **(Ex: 13052)** and **password** > Click **Submit** button or press **Enter** key



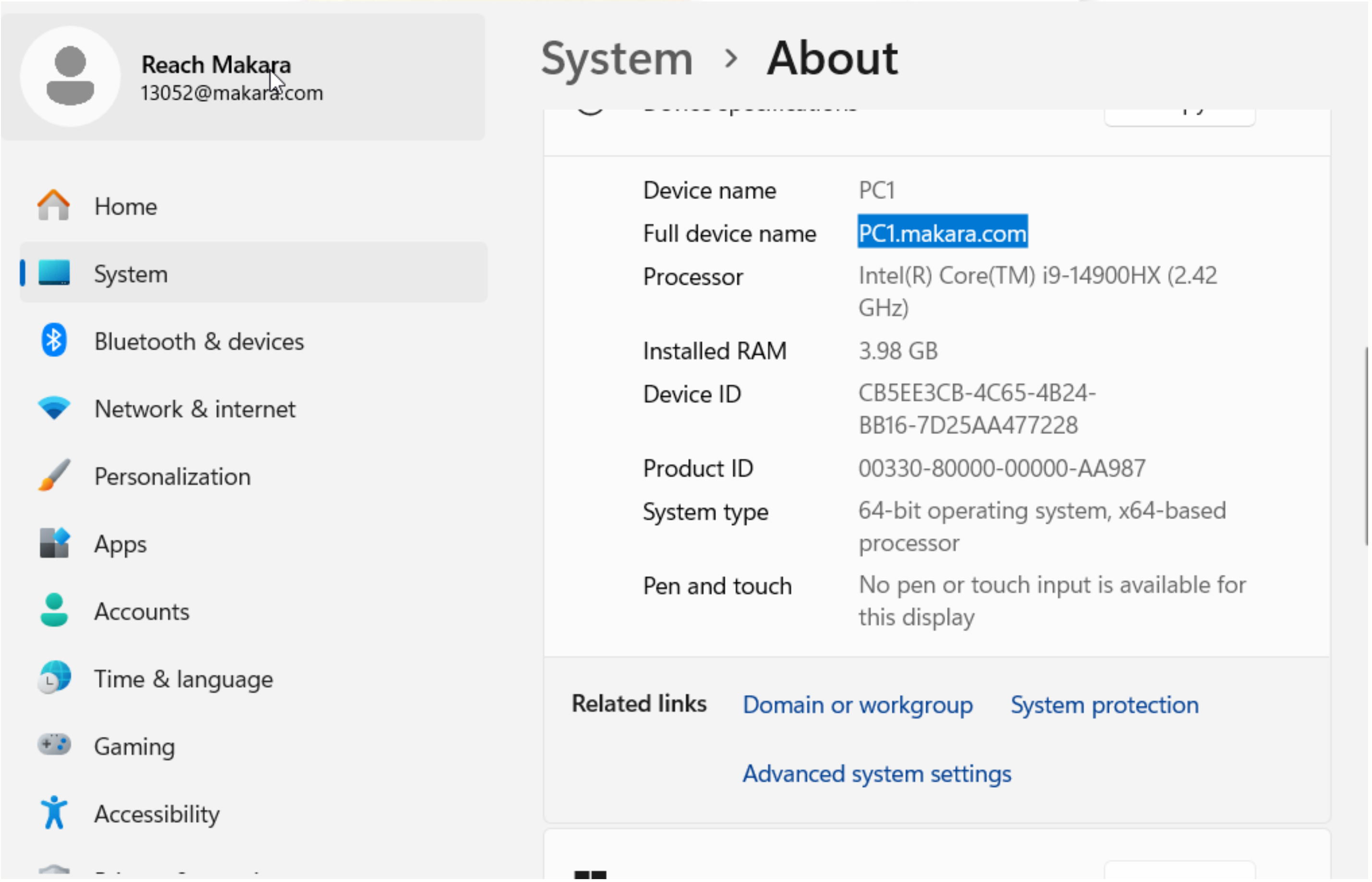
3.3. Join a Computer to a Domain (Cont.)

- Step 8: Press **Start + E key** to open **File Explorer** > Right click on **This PC** > Select **Properties**



3.3. Join a Computer to a Domain (Cont.)

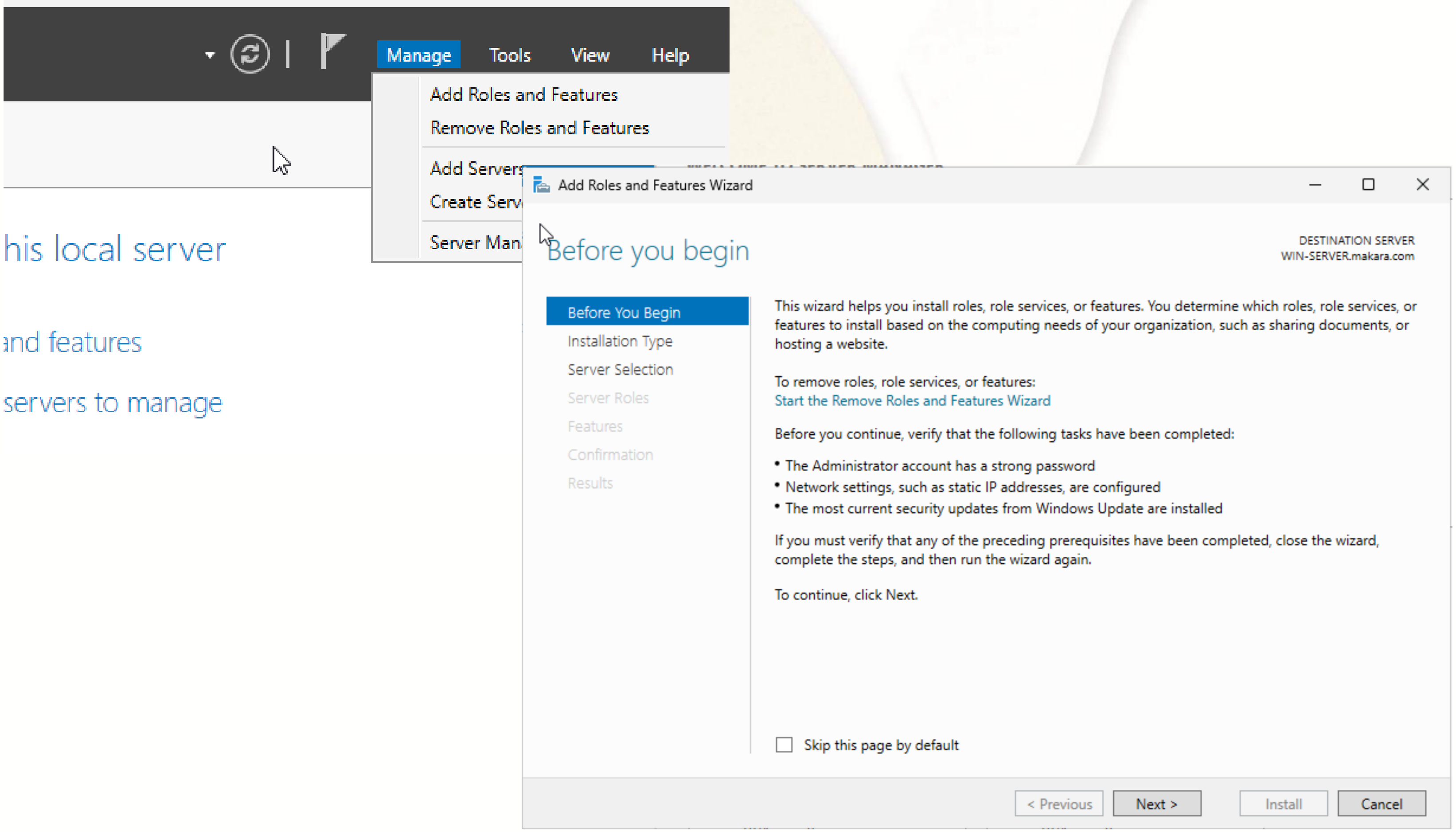
- Step 9: You will see this computer name with domain



4. Backup
Active Directory

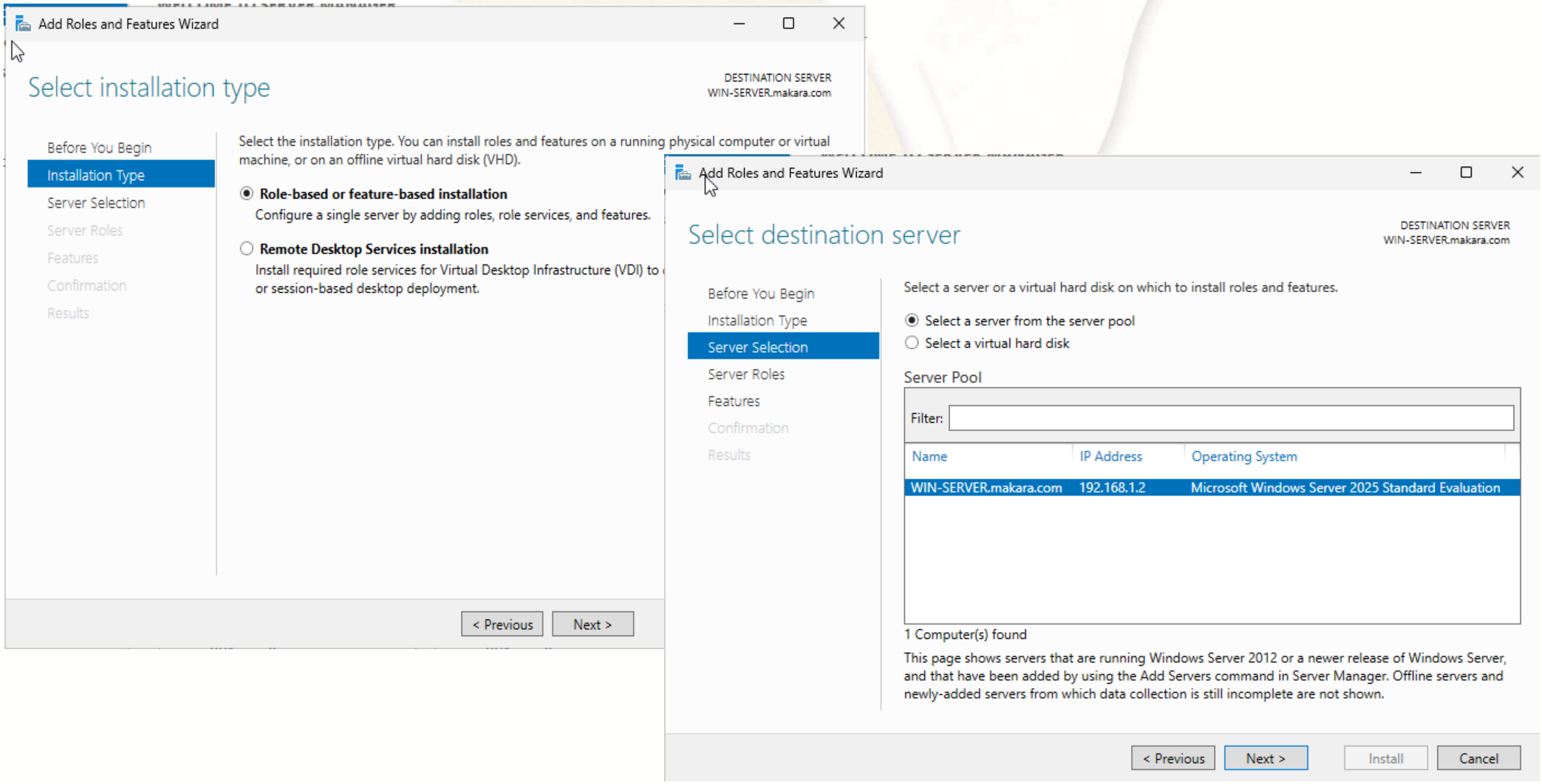
4.1. Install Windows Backup Feature

- Step 1: Click **Manage menu** > Click **Add Roles and Features** > Click **Next**



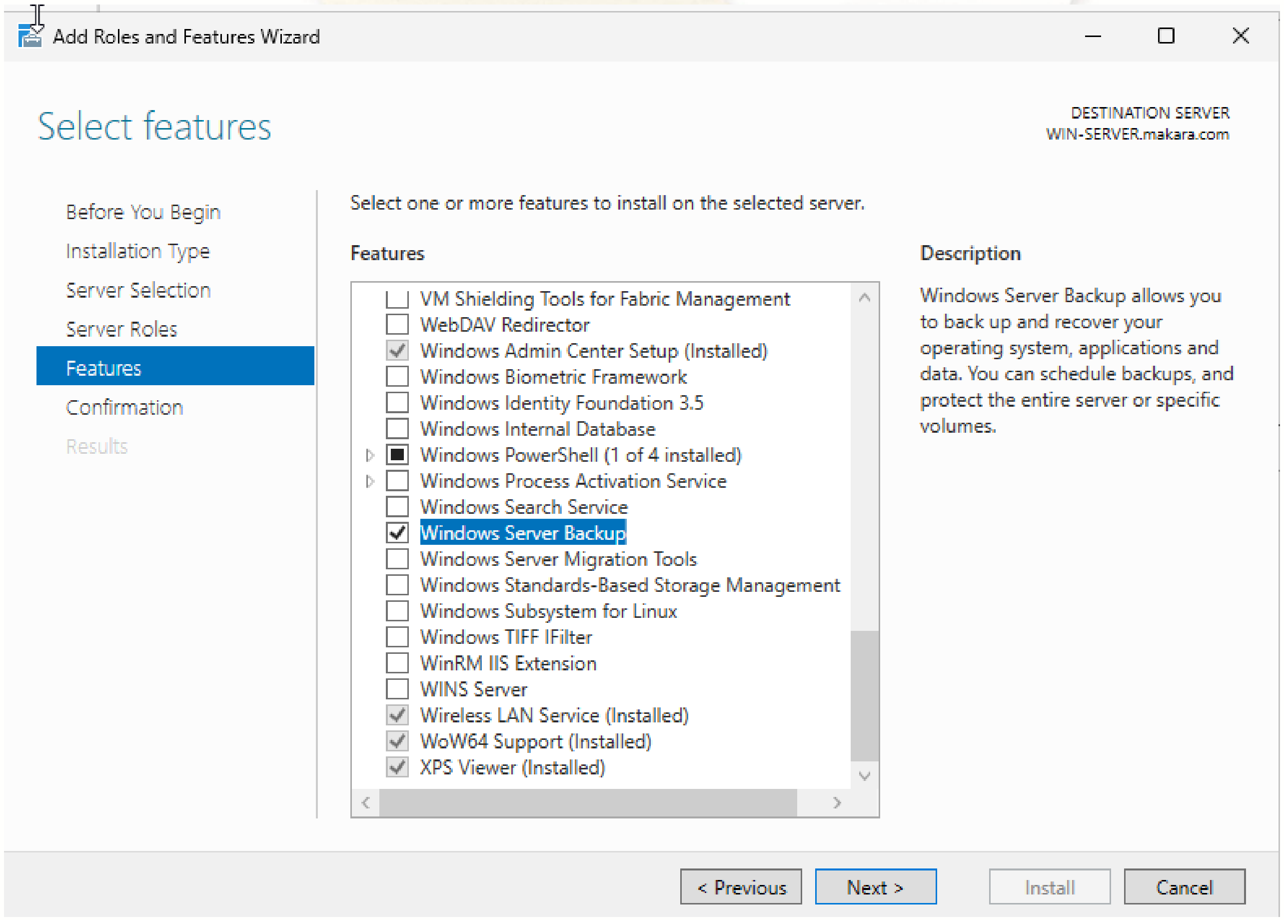
4.1. Install Windows Backup Feature (Cont.)

- Step 2: Click **Next** on **Select installation type** window > Click **Next**



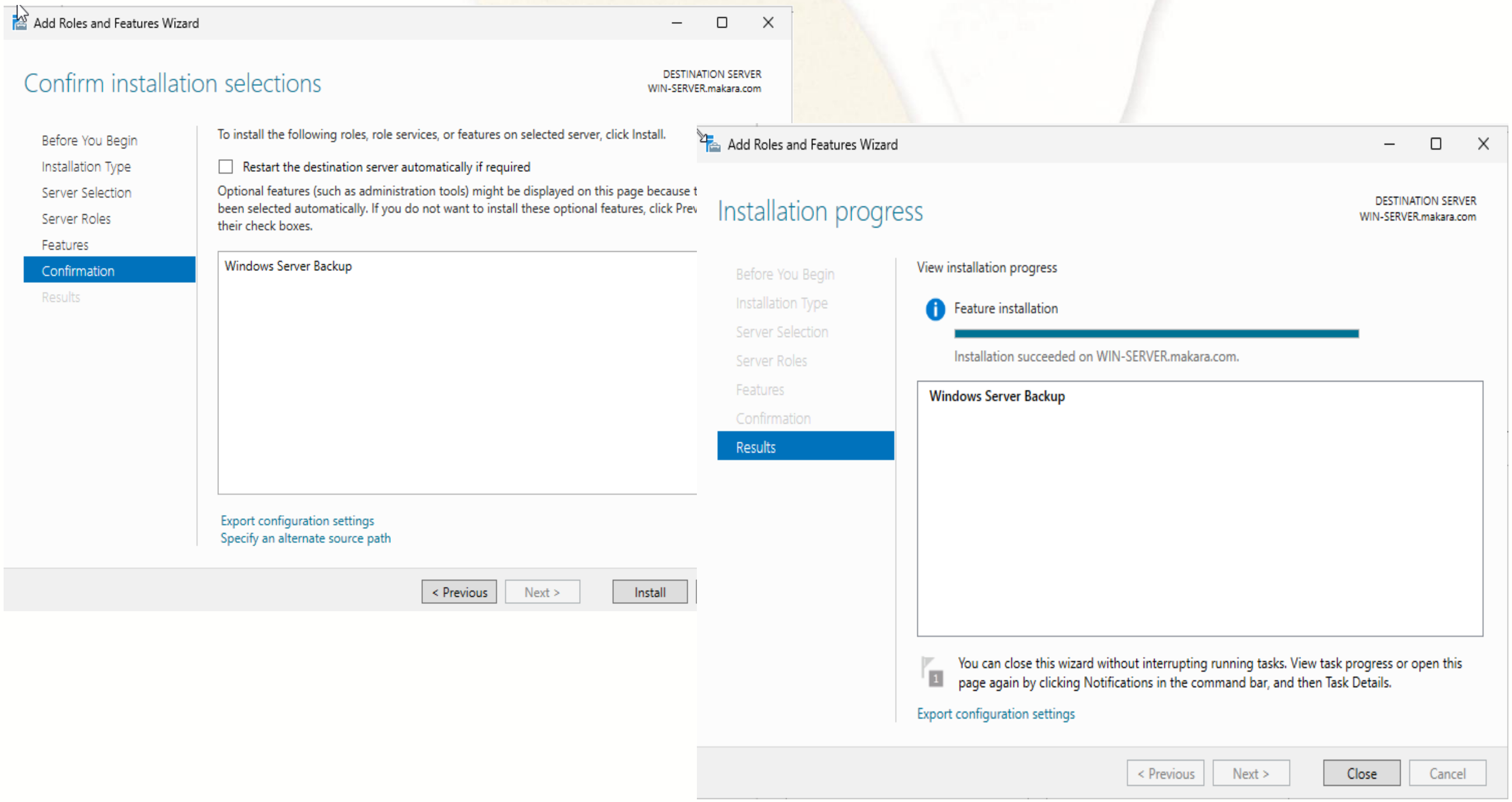
4.1. Install Windows Backup Feature (Cont.)

- Step 3: Select **Features** > Check **Windows Server Backup** > Click **Next**



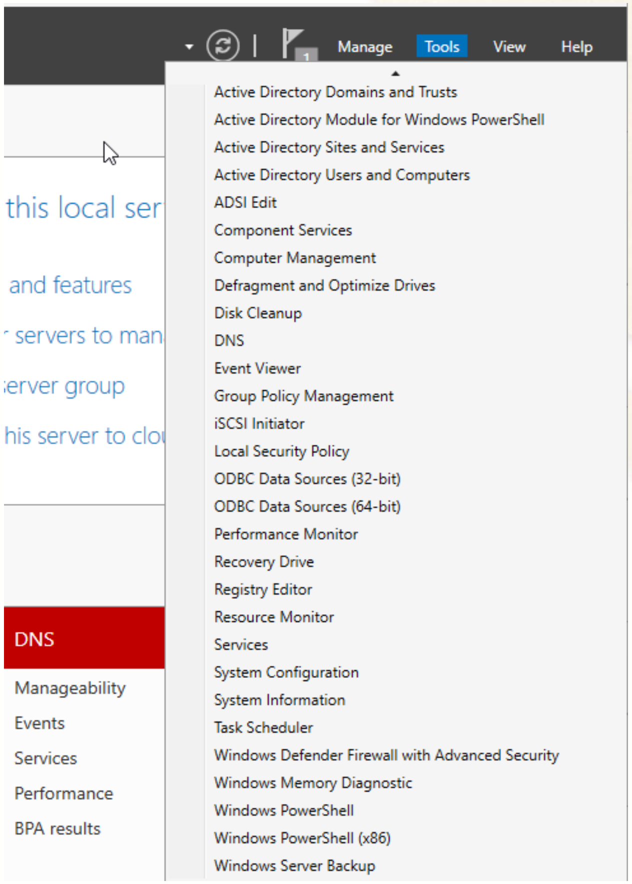
4.1. Install Windows Backup Feature (Cont.)

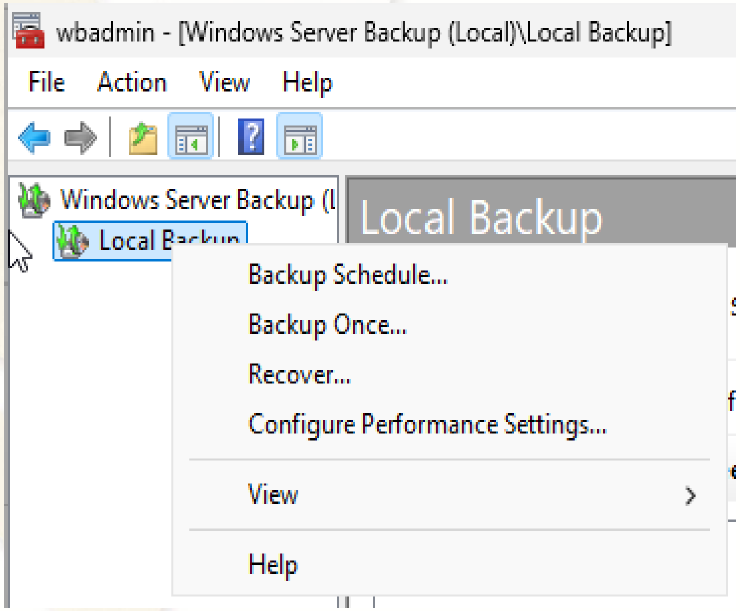
- Step 4: Click **Install** > wait **until installation succeeded**, click **Close** button



Backup Once AD

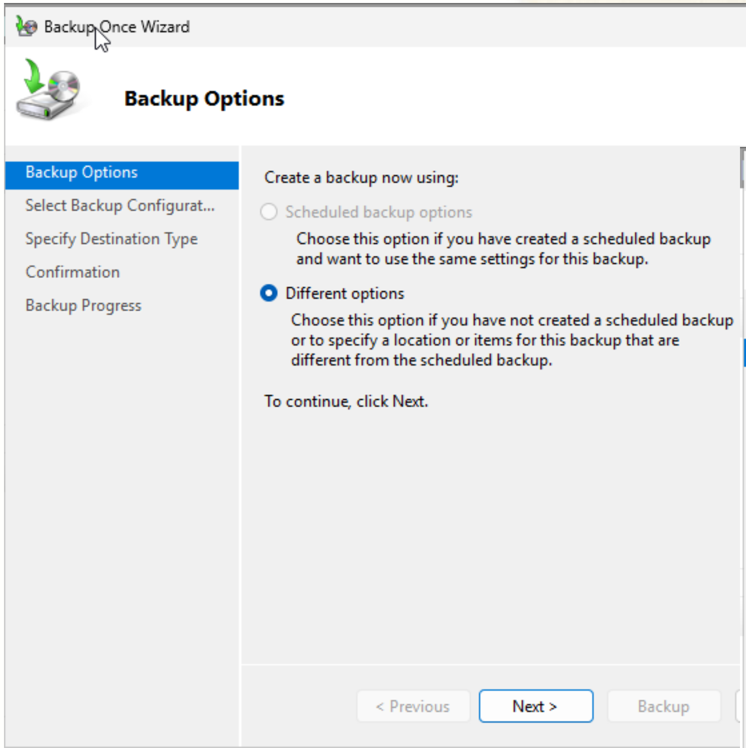
- Step 1: Click **Tools** menu and choose **Windows Server Backup** > Right click on **Local Backup** > Select **Backup Once...**

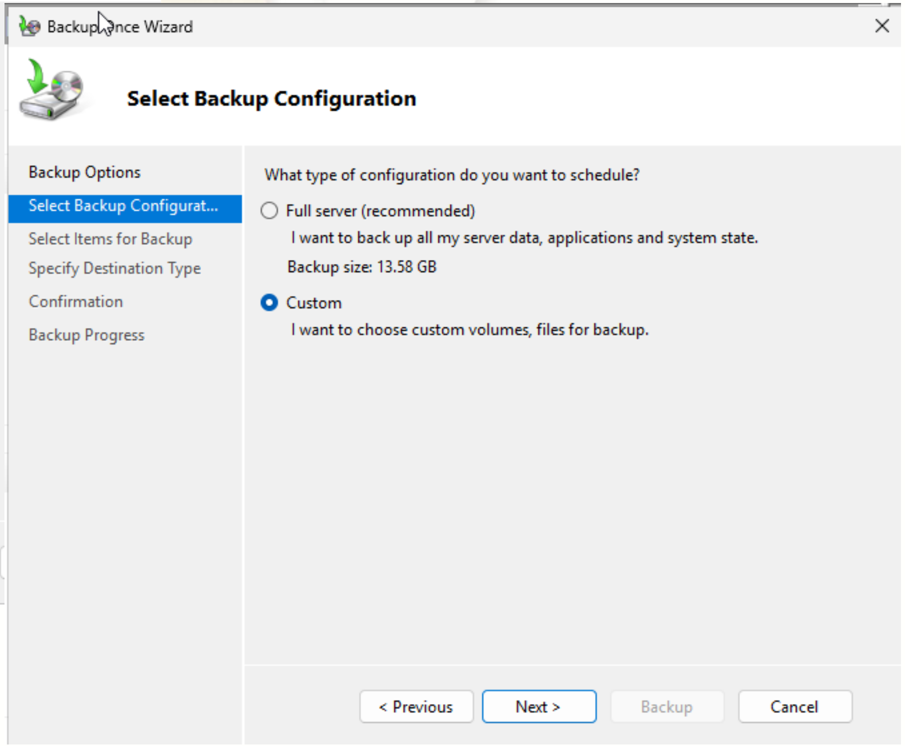




4.2. Backup Once AD (Cont.)

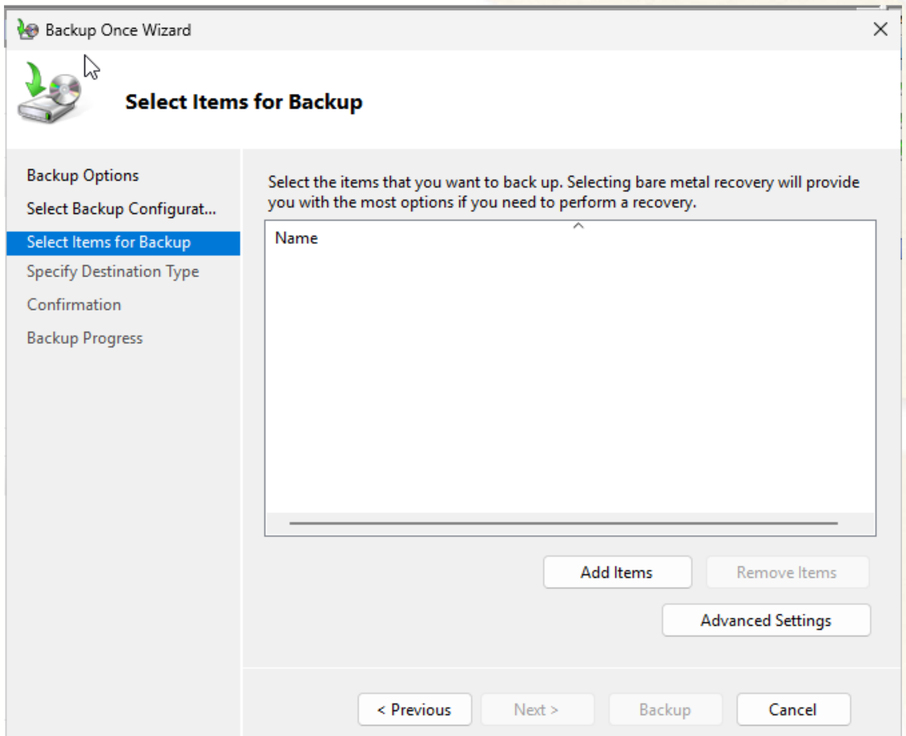
- Step 2: Select **Different options** > Click **Next** > Select **Custom** > Click **Next**

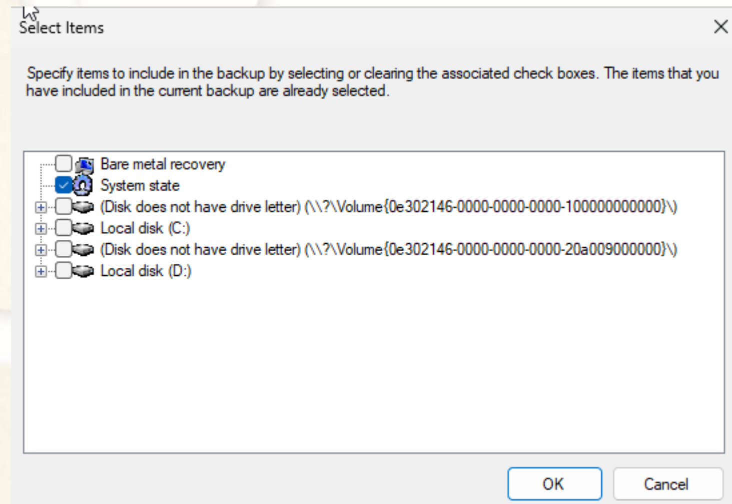




4.2. Backup Once AD (Cont.)

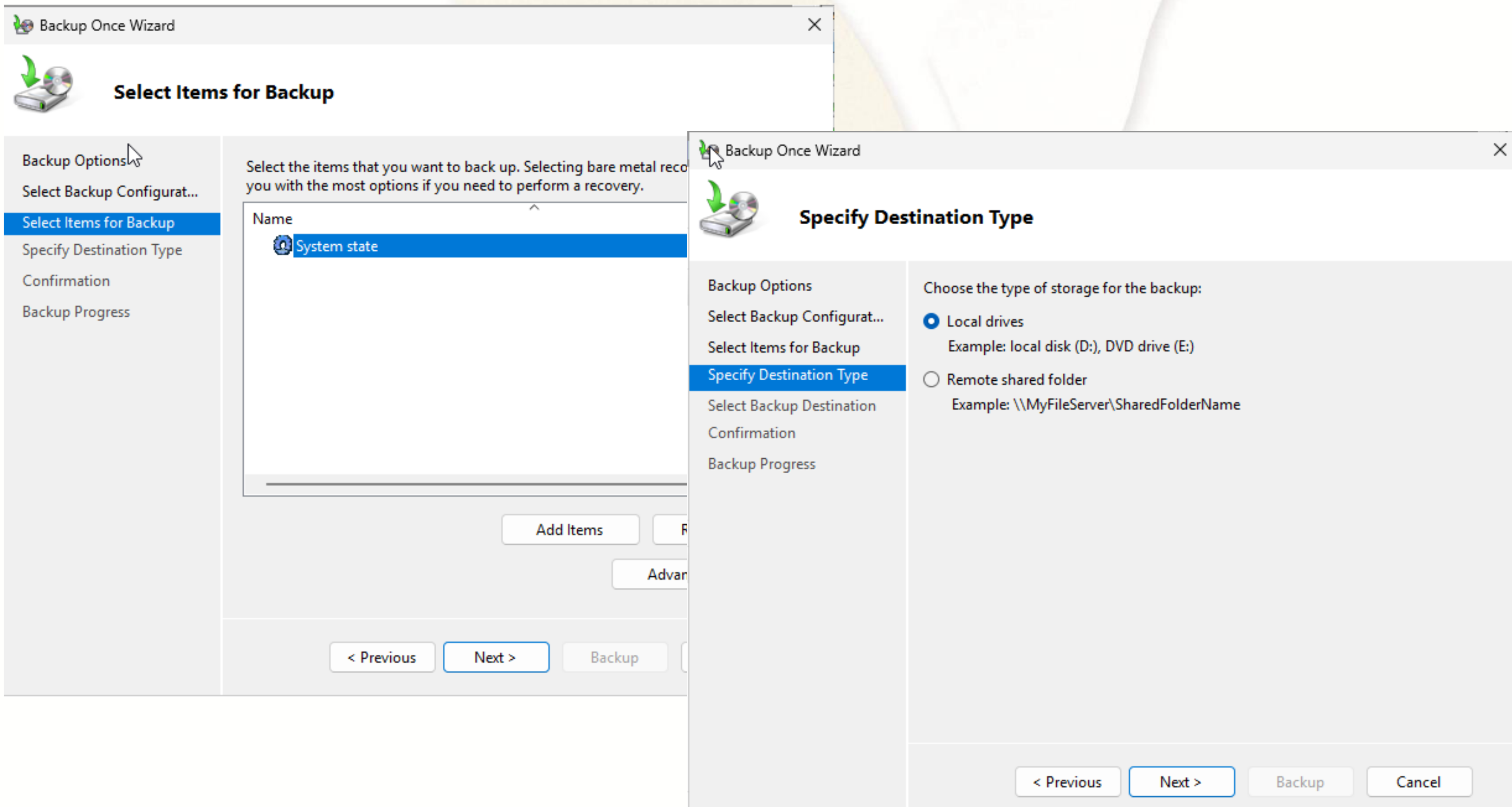
- Step 3: Click **Add Items** button > Choose **System state** > Click **OK**





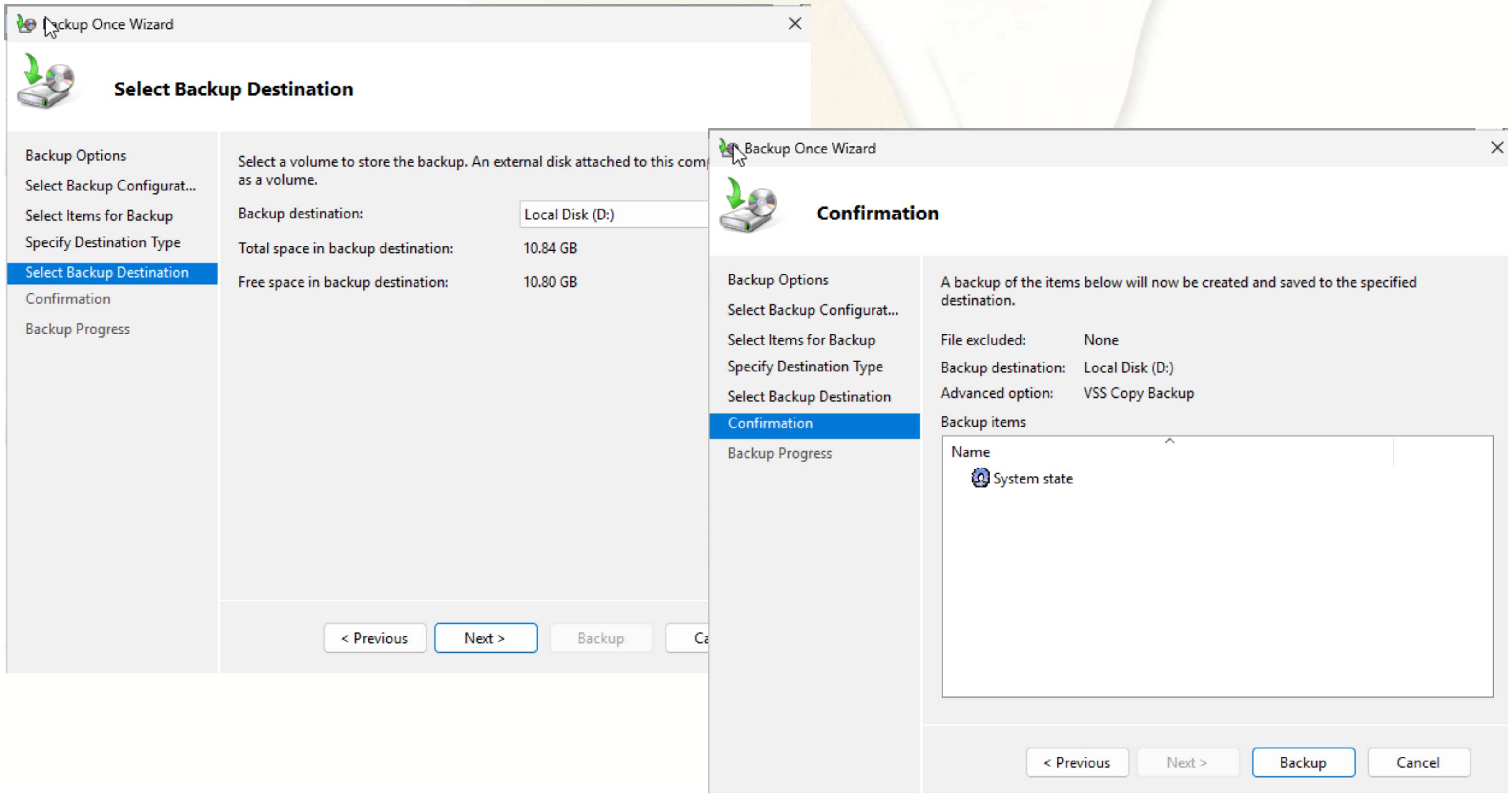
Backup Once AD (Cont.)

- Step 4: Click **Next** > Select **Local drives** > Click **Next**



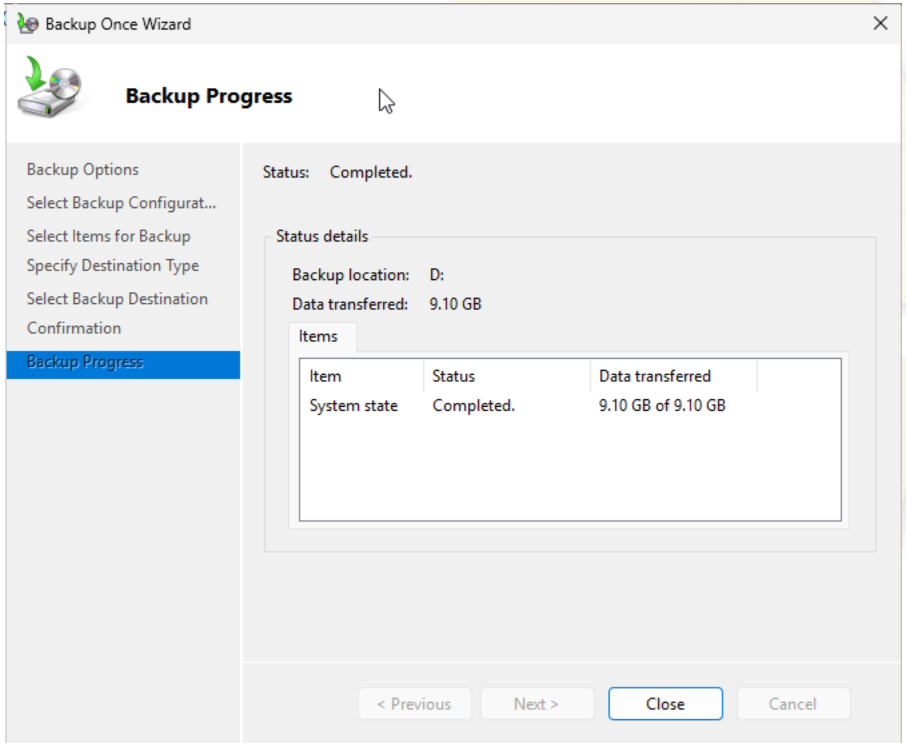
Backup Once AD (Cont.)

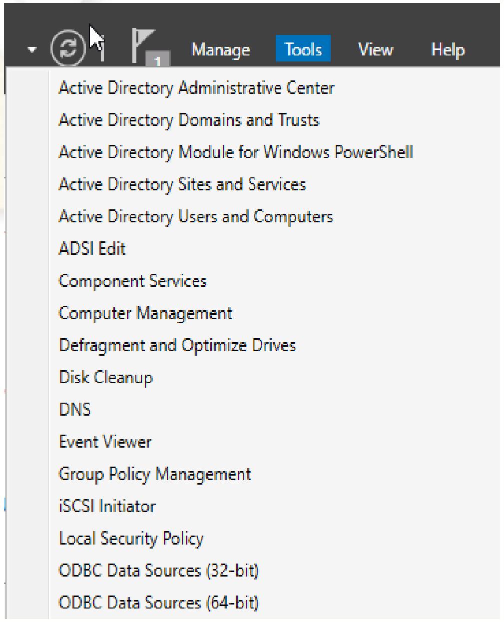
- Step 5: Select **Local Disk (D:)** or specific backup file location > Click **Next** > Click **Backup** button



4.2. Backup Once AD (Cont.)

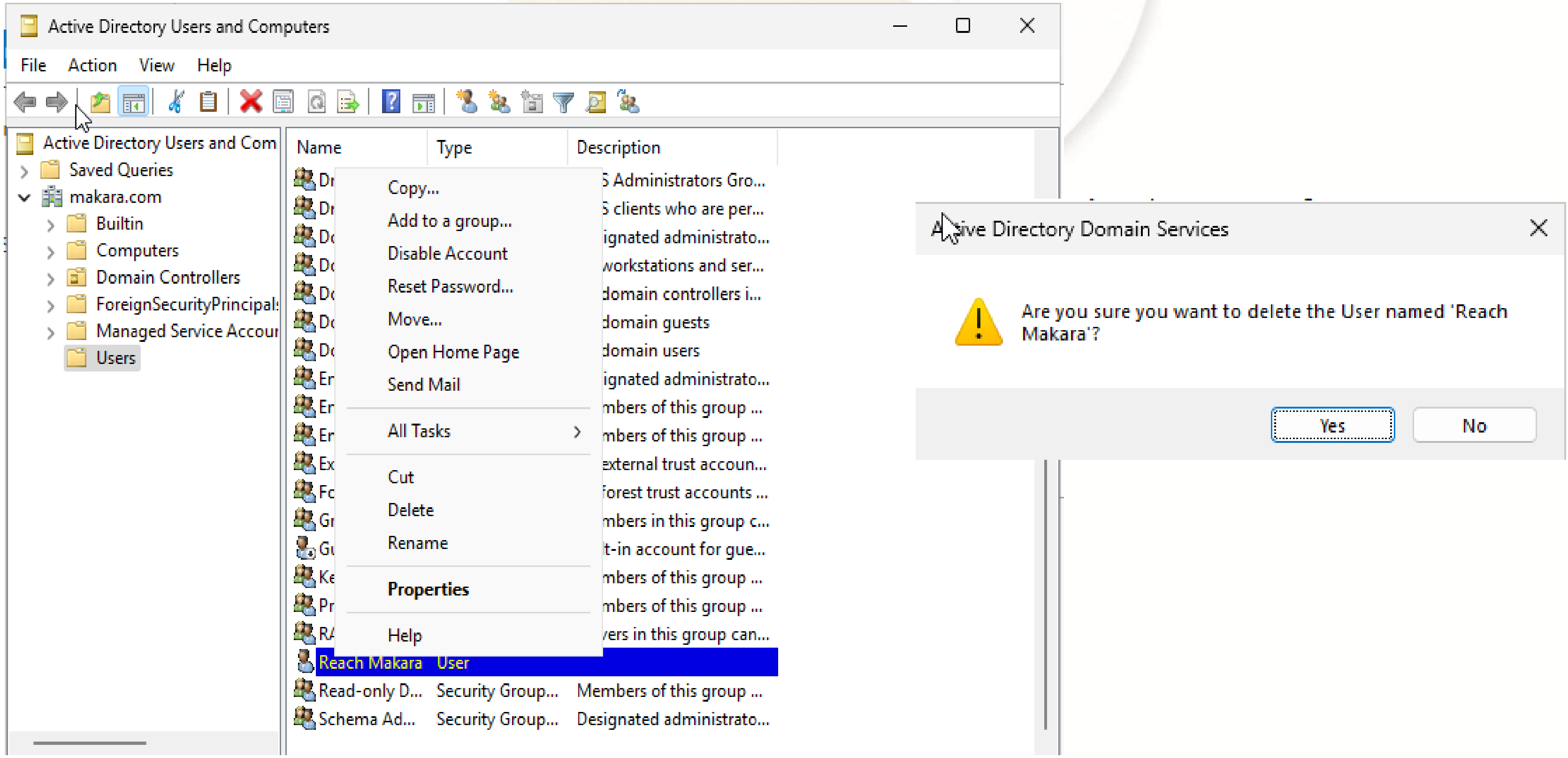
- Step 6: When **backup progress** has been **completed**, Click **Close** > Go to **Server Manager** and Click **Tools menu** > Click **Active Directory Users and Computers**





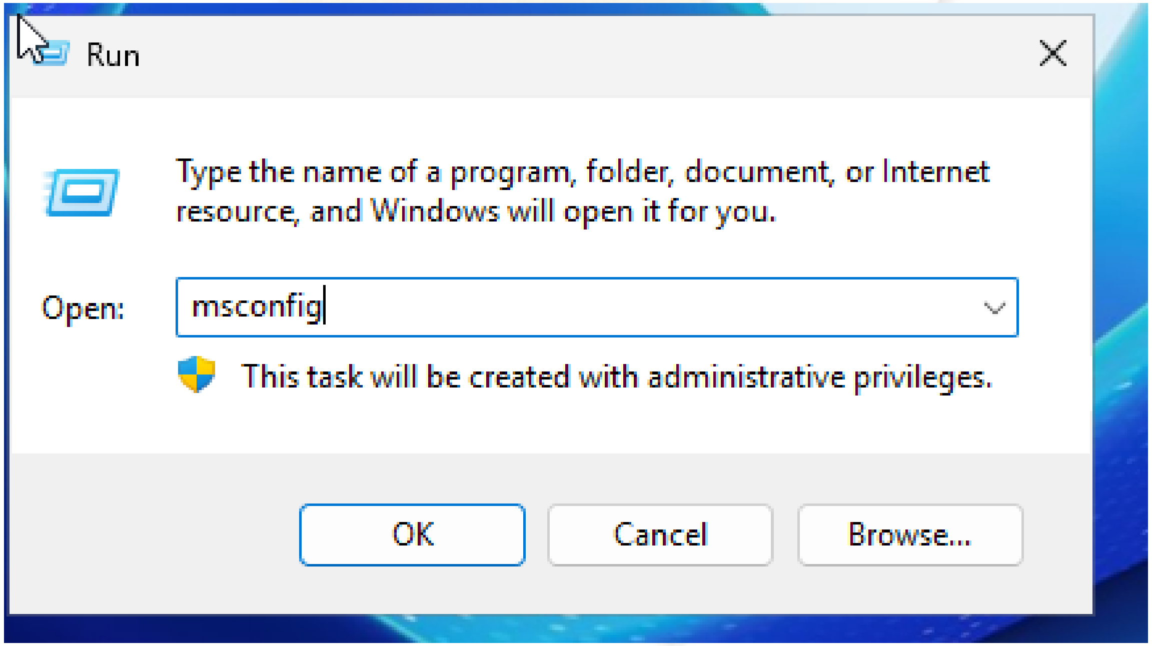
4.2. Backup Once AD (Cont.)

- Step 7: Right Click on a **user (Reach Makara)** > Select **Delete** > Click **Yes** button **(this user will restore after recovery process)**



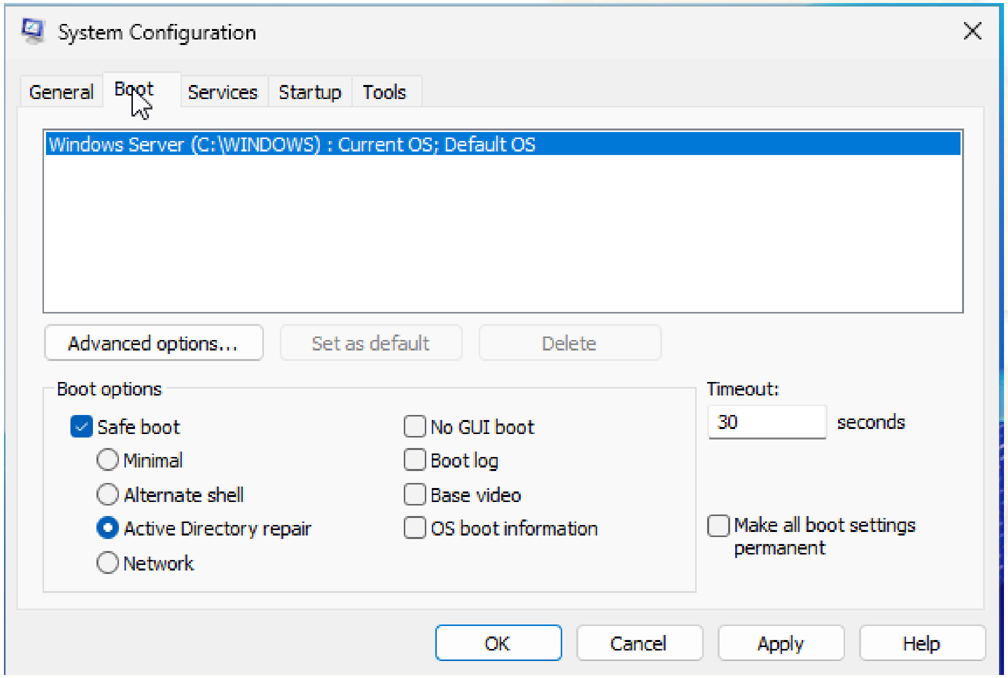
4.3. Restore AD

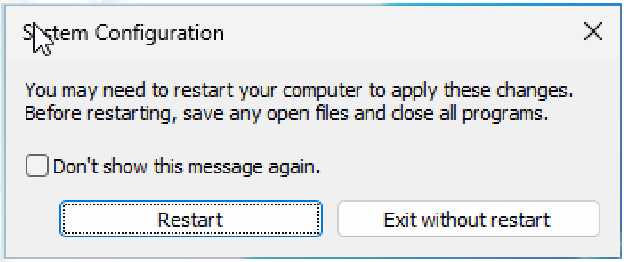
- Step 1: Go to **Run** by key **Windows + R** > Type **msconfig** and Click **OK**



4.3. Restore AD (Cont.)

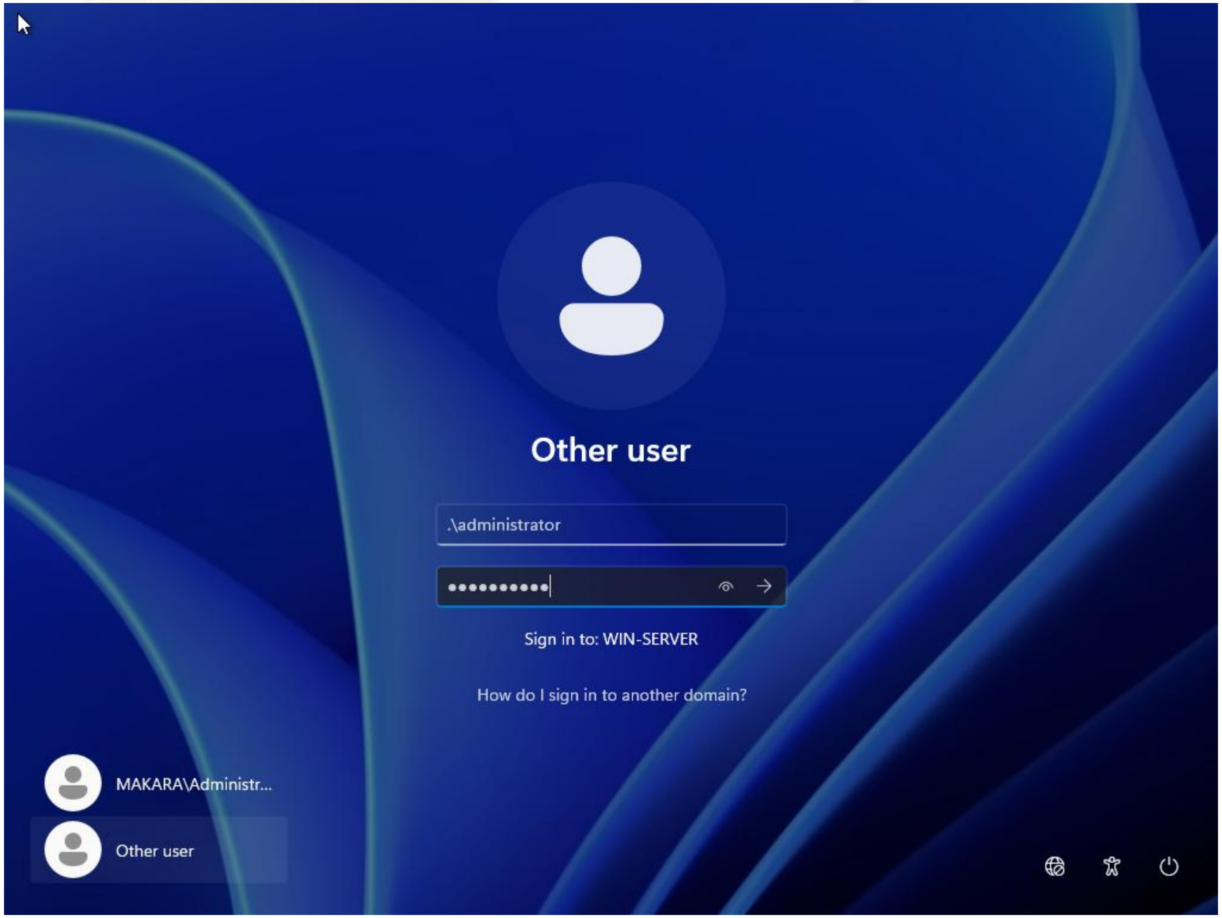
- Step 2: Select **Boot** tab > Check **Safe boot** > Select **Active Directory repair** > Click **OK** > Click **Restart**





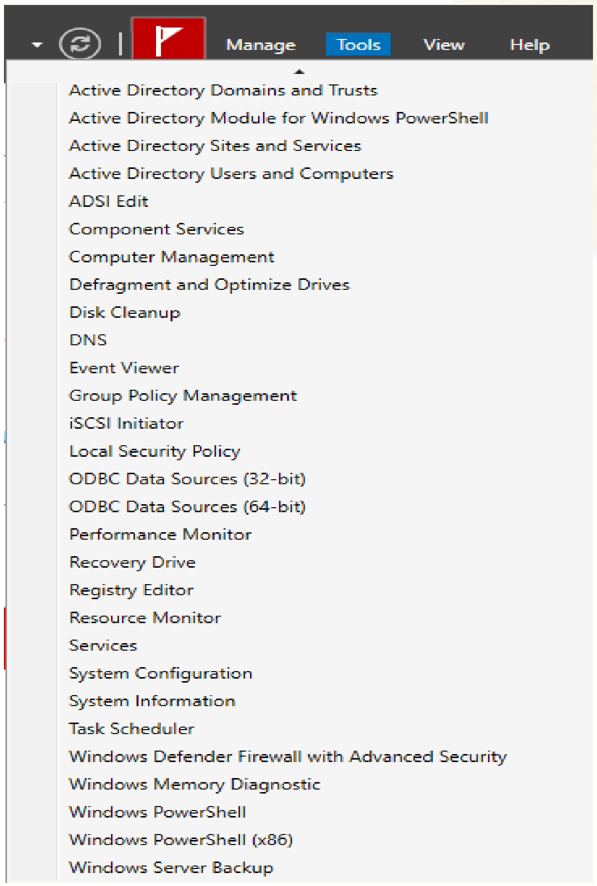
4.3. Restore AD (Cont.)

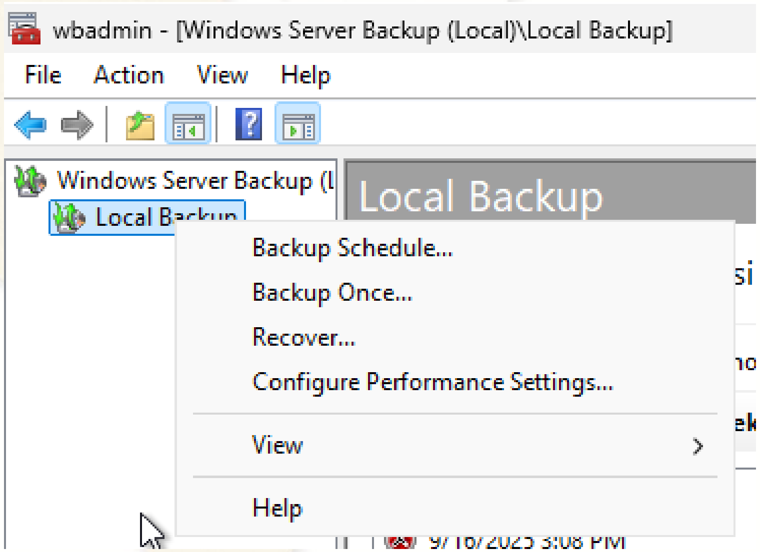
- Step 3: Click **Other user** > Enter **local account** information such as **username** (.\Administrator) and **password**



4.3. Restore AD (Cont.)

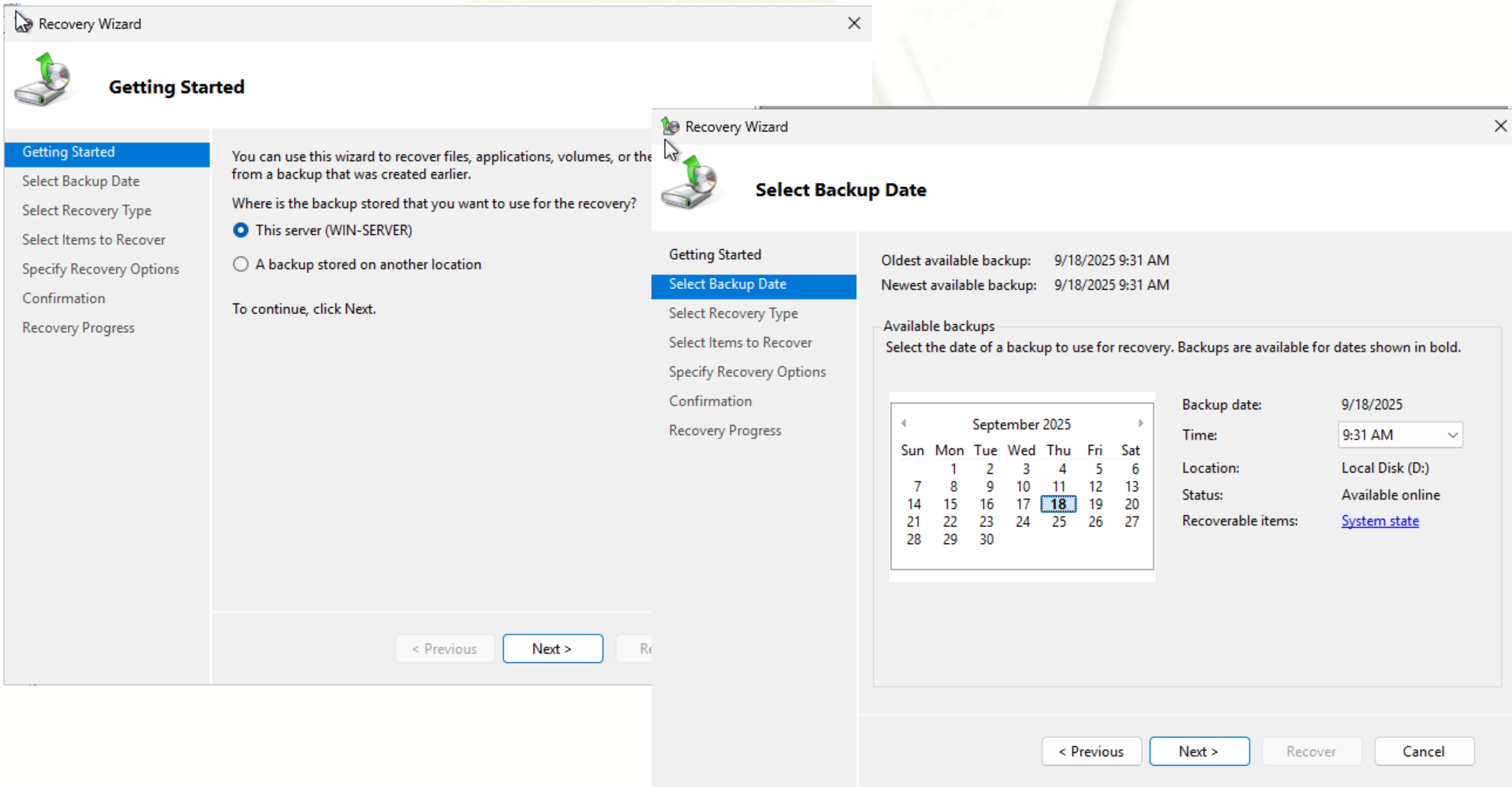
- Step 4: Go to **Server Manager** and Click **Tools** menu > Select **Windows Server Backup** > Right click on **Local Backup** > Select **Recover…**





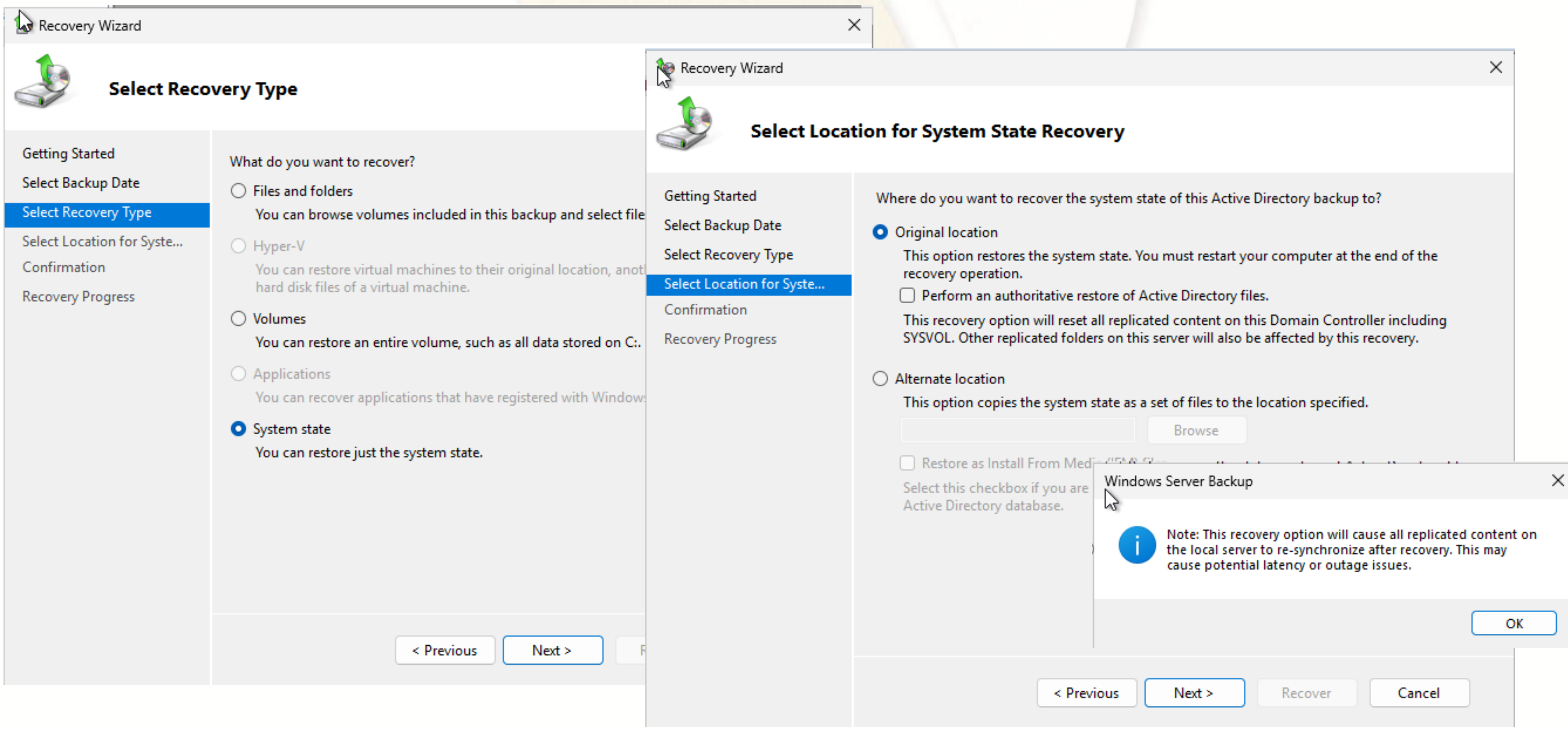
4.3. Restore AD (Cont.)

- Step 5: Select **This server** > Click **Next** > Select recent **backup date** > Click **Next**



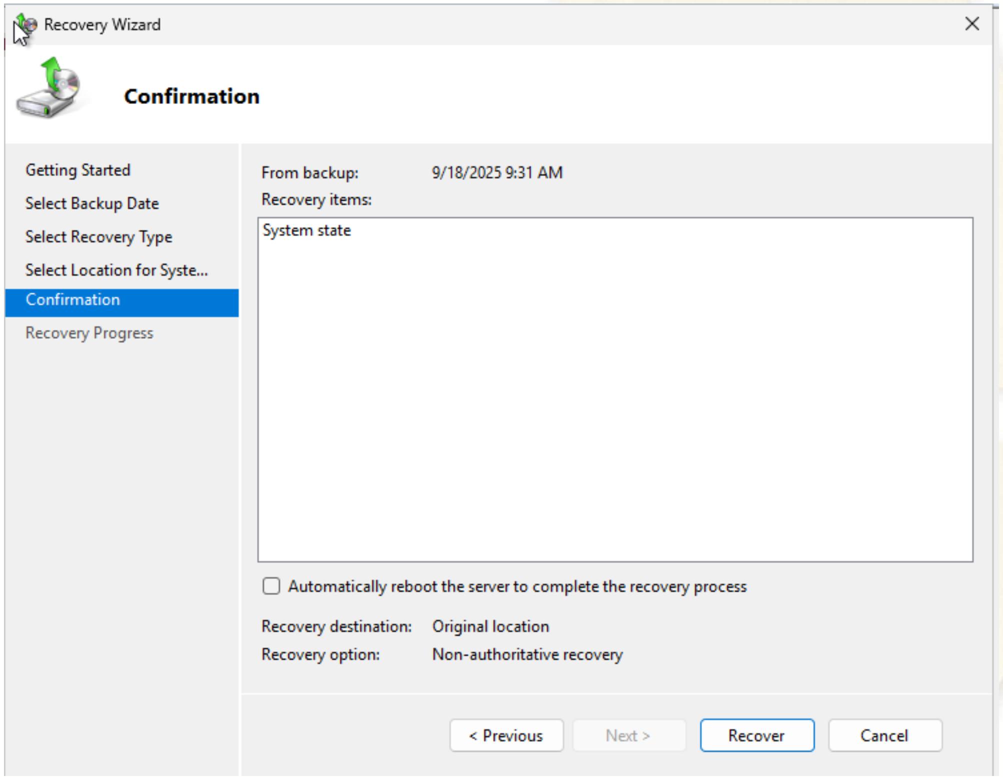
4.3. Restore AD (Cont.)

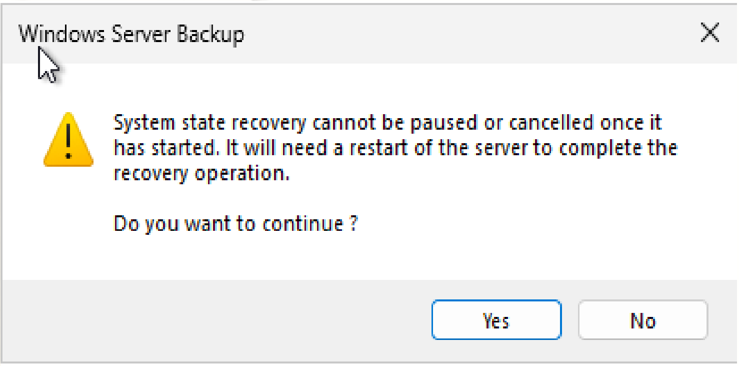
- Step 7: Select **System state** > Click **Next** > Select **Original location** > Click **Next** > Click **OK**

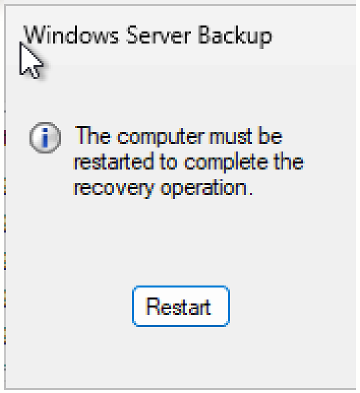


4.3. Restore AD (Cont.)

- Step 8: Click **Recover** > Click **Yes** button > Wait until **recovery process** has been **completed**, click **Restart** button

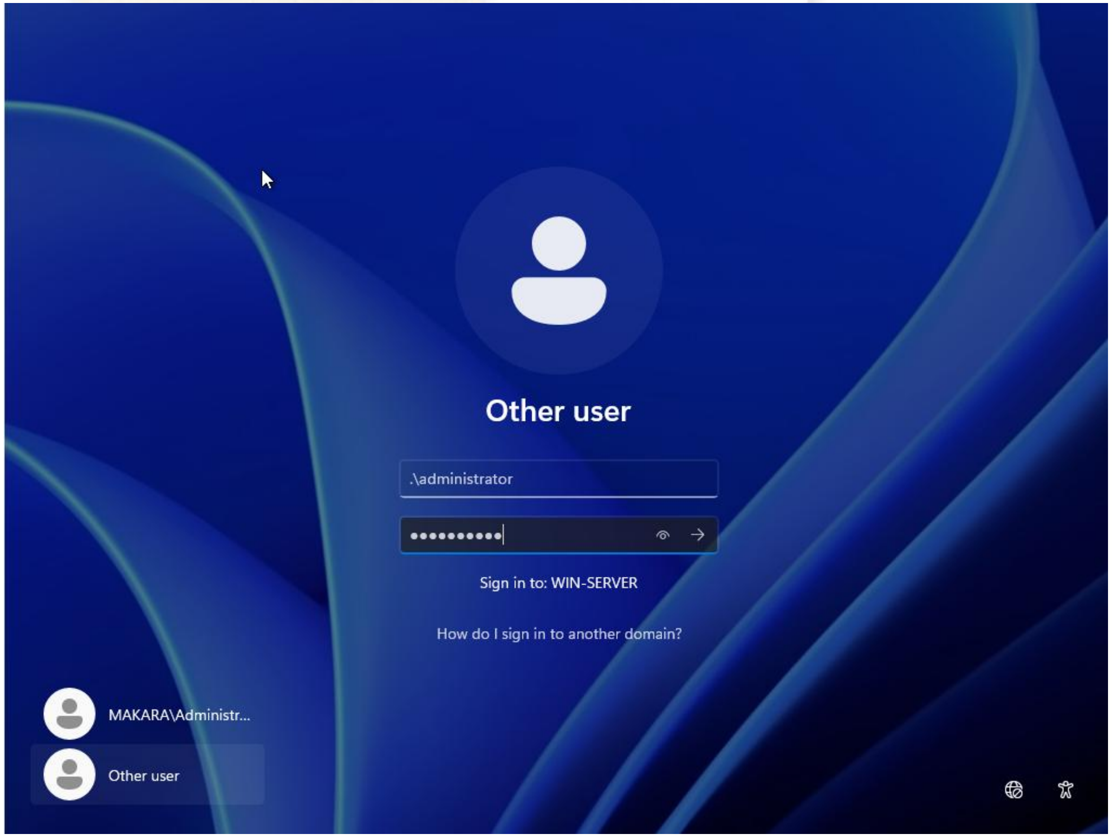






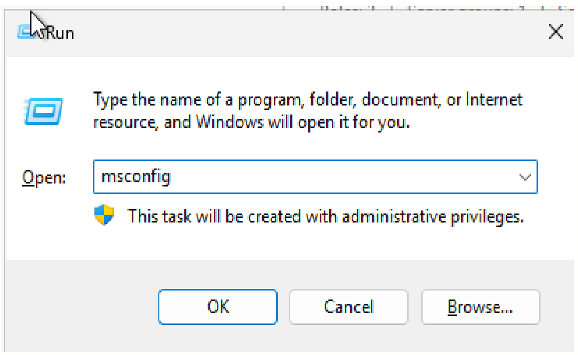
4.3. Restore AD (Cont.)

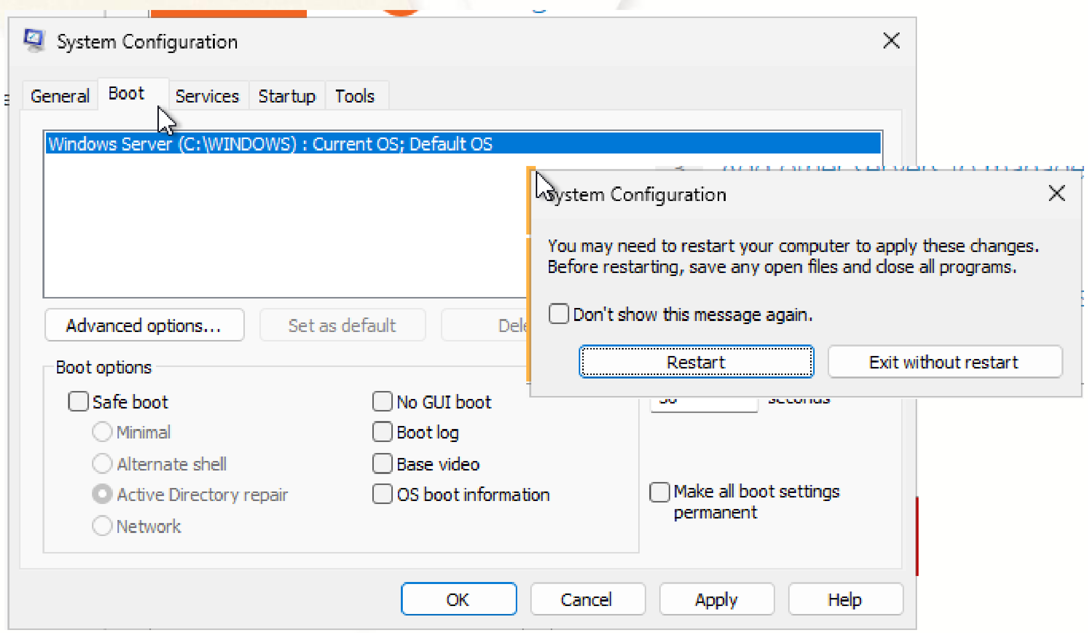
- Step 9: Click **<u>Other user</u>** > Enter **local account** information such as **<u>username</u>** (.\Administrator) and **<u>password</u>**



4.3. Restore AD (Cont.)

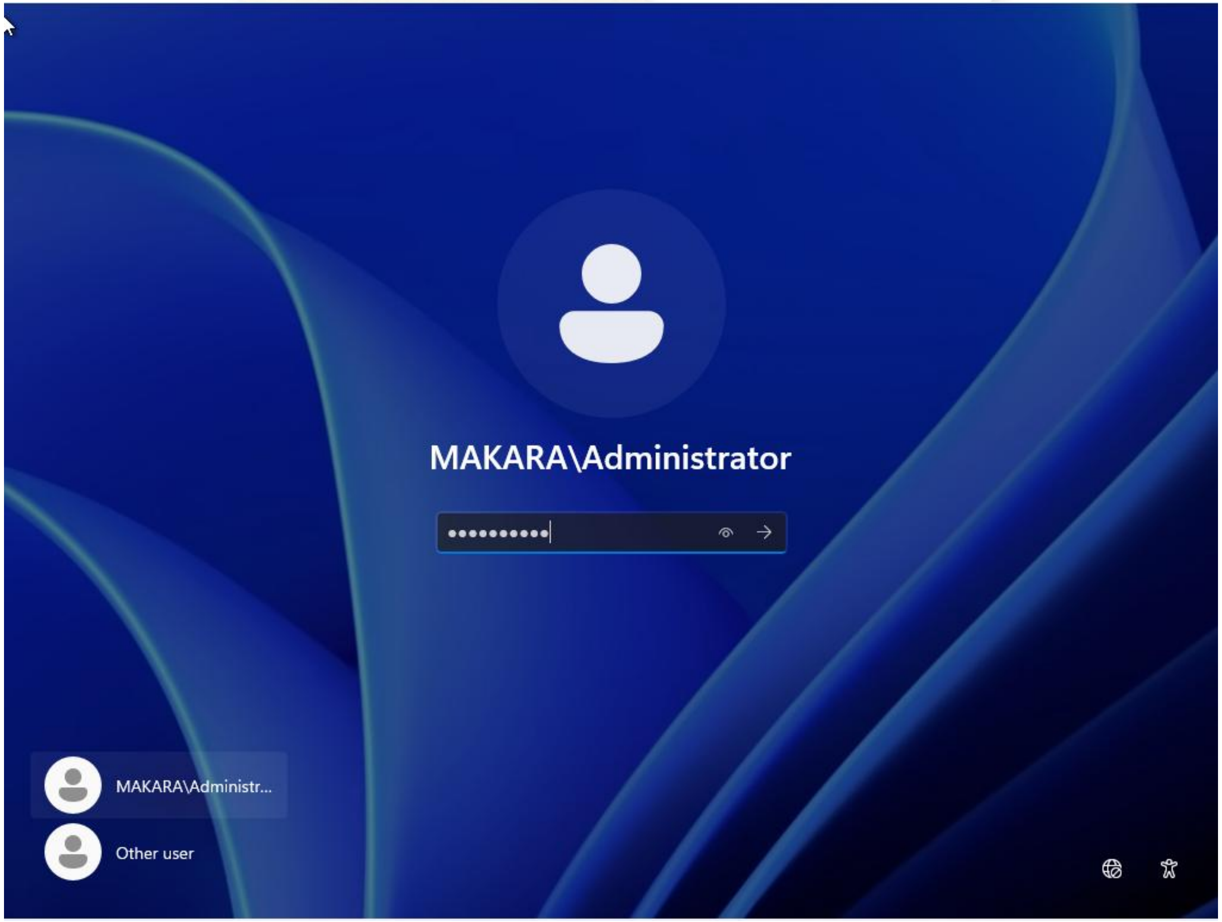
- Step 10: **Start + R** key to open **Run** Windows > Type **msconfig** and Click **OK** > Select **Boot** tab > Uncheck **Safe boot** > Click **OK** > Click **Restart**





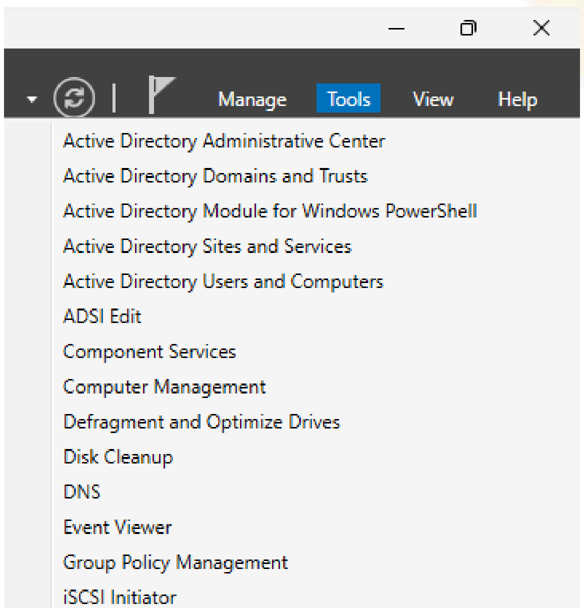
4.3. Restore AD (Cont.)

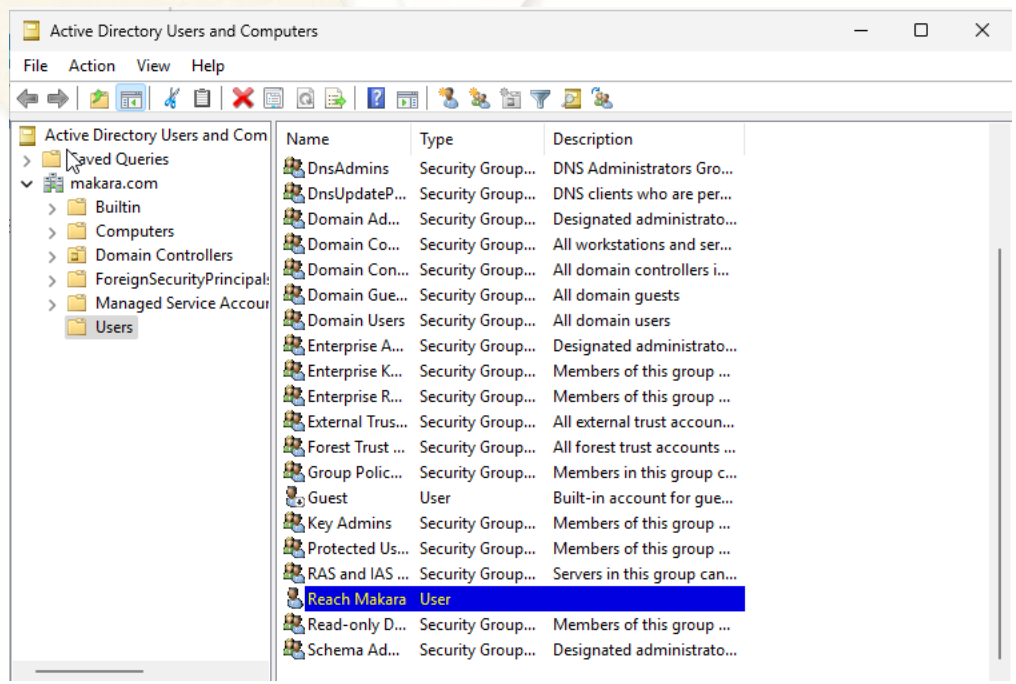
- Step 11: Click on **MAKARA\Administrator** > hit **password** and Enter for **Sign in**



4.3. Restore AD (Cont.)

- Step 12: Go to **Server Manager** and Click **Tools** tab > Click **Active Directory Users and Computers** > Go to **Users Folder** and in **user list** you will see **Reach Makara** has been **recovered**





Thank You.